Gut Microbiome Signature Correlation Analysis

Pairwise Comparison Report

Reproduction (Phase 1) + extension to non−bariatric GI surgery (Phase 2)

+ bariatric positive control, RYGB−specific and non−RYGB sub−phases (Phase 3)

All Datasets Used In This Analysis

12 cohorts across 3 phases, all 16S rRNA, all repeat–sampled pre/post surgery

Cohort	Phase	Accession	Procedure	Subjects	Runs
BS	1 (OG)	PRJNA668472	RYGB	30	61
Assal	1 (OG)	PRJNA395660	RYGB	25	71
Afshar	1 (OG)	PRJEB14082	RYGB	31	50
Ilhan	1 (OG)	PRJNA612042	RYGB	45	82
Cholecystectomy	2	PRJNA1188648	Gallbladder removal	10	30
IleocecalResection	2	PRJNA514452	Ileocecal resection (Crohn's)	54	189
Ileostomy	2	PRJNA1480144 (IL)	Temporary ileostomy	44	65
SDT	2	PRJNA1480144 (SD)	Stent-based fecal diversion	34	45
RYGB_SaoPaulo	3b	PRJNA1364303	RYGB +/- exercise	31	87
Lazaro_GB	3b	PRJNA859272 (GB)	RYGB	10	20
Lazaro_SG	3a	PRJNA859272 (SG)	Sleeve gastrectomy	14	28
Chaudhari_SG	3a	PRJNA635168	Sleeve gastrectomy	17	34

Phase 1 = original paper's 4 cohorts ("OG"). Phase 2 = unrelated GI surgeries, tests whether the signature generalizes to any gut surgery. Phase 3 = more bariatric cohorts as a positive control, split into RYGB–specific (3b) and non–RYGB bariatric / sleeve gastrectomy (3a).

How the Heatmaps and Scatterplots Were Built

In plain terms, no statistics background required

1. For every cohort, each subject's gut bacteria were measured before surgery and at one or more points after. Each distinct type of bacteria (a "taxon", e.g. the genus Veillonella) gets its own abundance reading per sample.

2. For each taxon in each cohort, a statistical test asks: did this bacterium's abundance change significantly after surgery? That test returns two things -- a p-value (how confident we are it's a real change, not noise) and a direction (did it go up or down).

3. Those two numbers are combined into one score per taxon: a large positive score means "significantly increased after surgery"; a large negative score means "significantly decreased"; near zero means "no real change detected."

4. SCATTERPLOT: to compare two cohorts, take every taxon they both measured and plot one cohort's scores on the x-axis against the other's on the y-axis. Each dot is one taxon. If the same bacteria rise and fall together in both cohorts, the dots trend along a diagonal line. If the two cohorts have nothing in common, the dots scatter randomly.

5. The tightness of that diagonal trend is summarized as a single number, the Spearman correlation (r), printed above each scatterplot along with a p-value for r itself. r near 1 = strong agreement between cohorts; r near 0 = no agreement.

6. HEATMAP: rather than looking at one scatterplot pair at a time, the heatmap puts every cohort/timepoint against every other in one grid. Each cell is just that pair's r and p value from step 5 -- the heatmap doesn't compute anything new, it's a compact way to see every pairwise comparison at once. Warmer colors (red) = stronger agreement, cooler colors (blue) = weaker. Grey cells compare a cohort's own two timepoints to each other (e.g. 1-month vs. 6-month within the same study) and are excluded from headline numbers, since the real test is agreement BETWEEN independent cohorts, not within one.

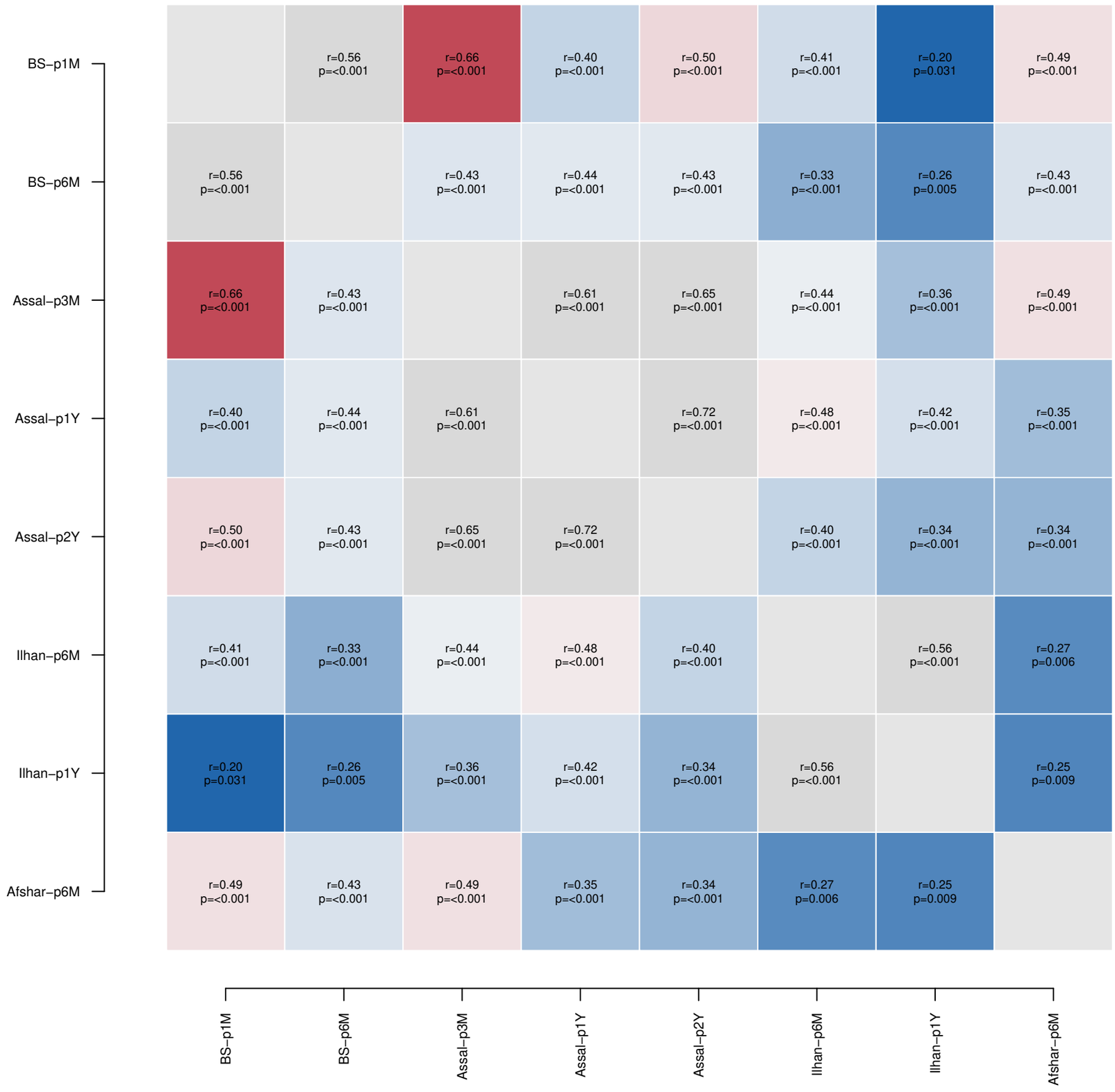
The headline number reported for each phase (e.g. "r = 0.40") is just the average of all the DIFFERENT-cohort cells in that phase's heatmap.

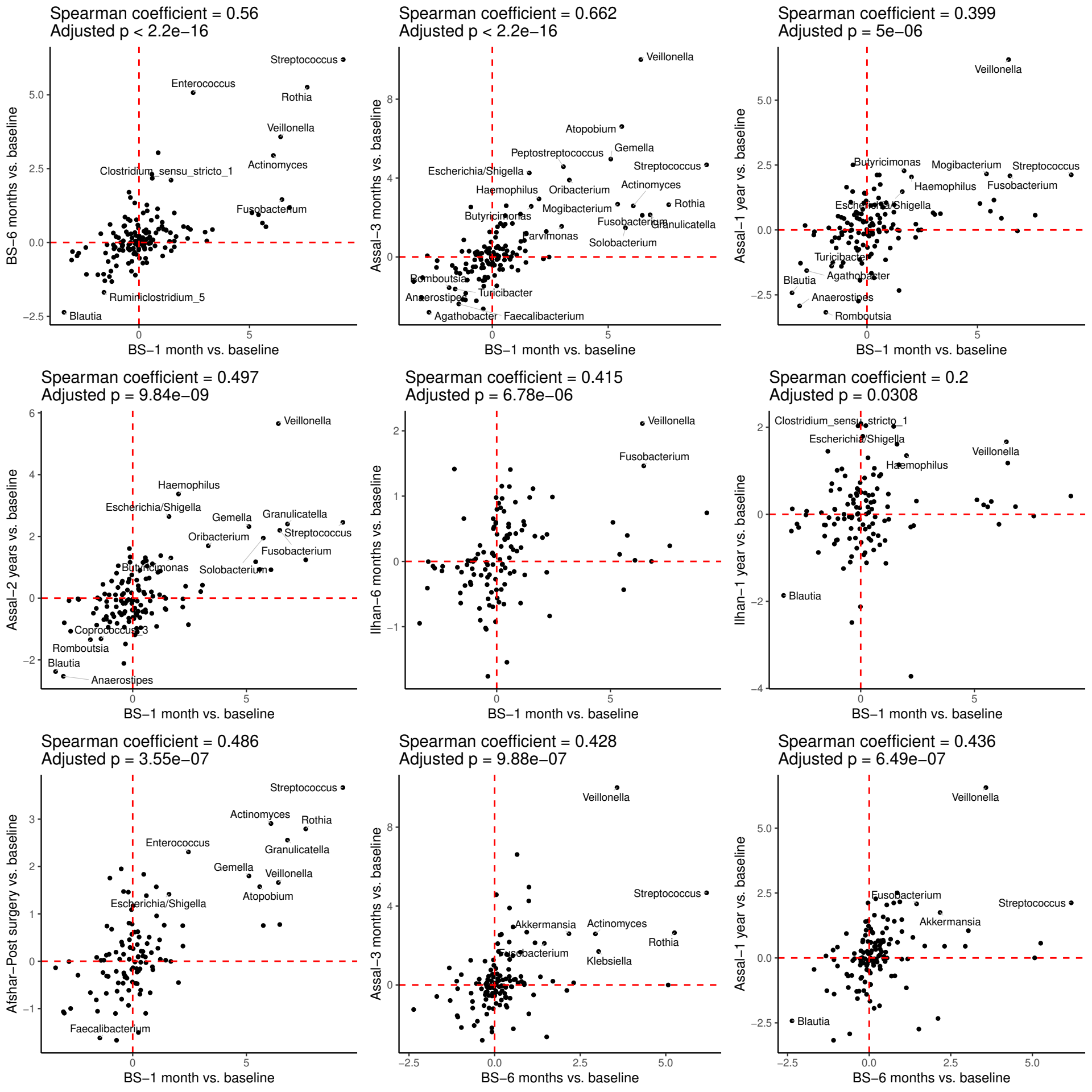
Phase 1: Reproduction

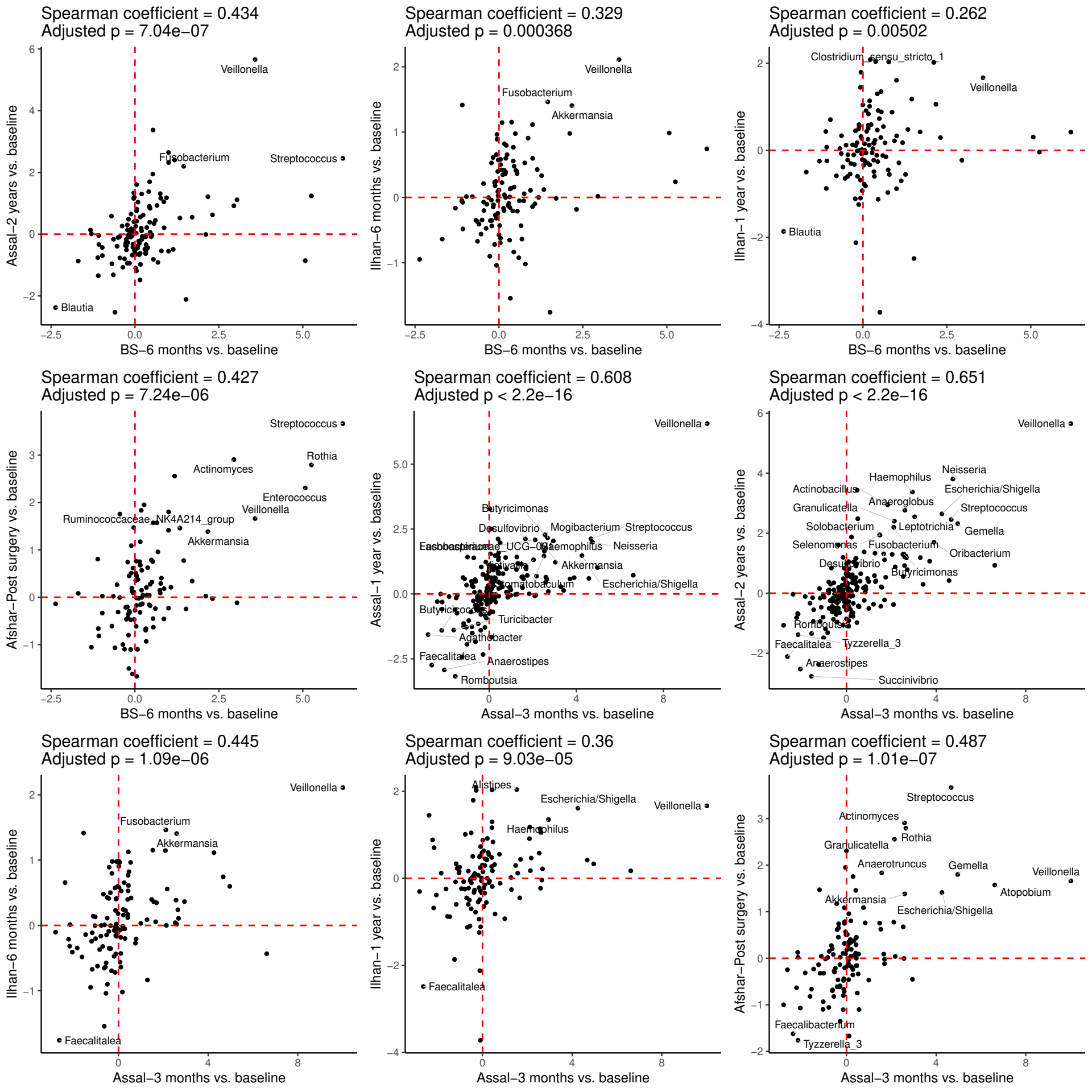
Cohorts: BS, Assal, Ilhan, Afshar (all RYGB)

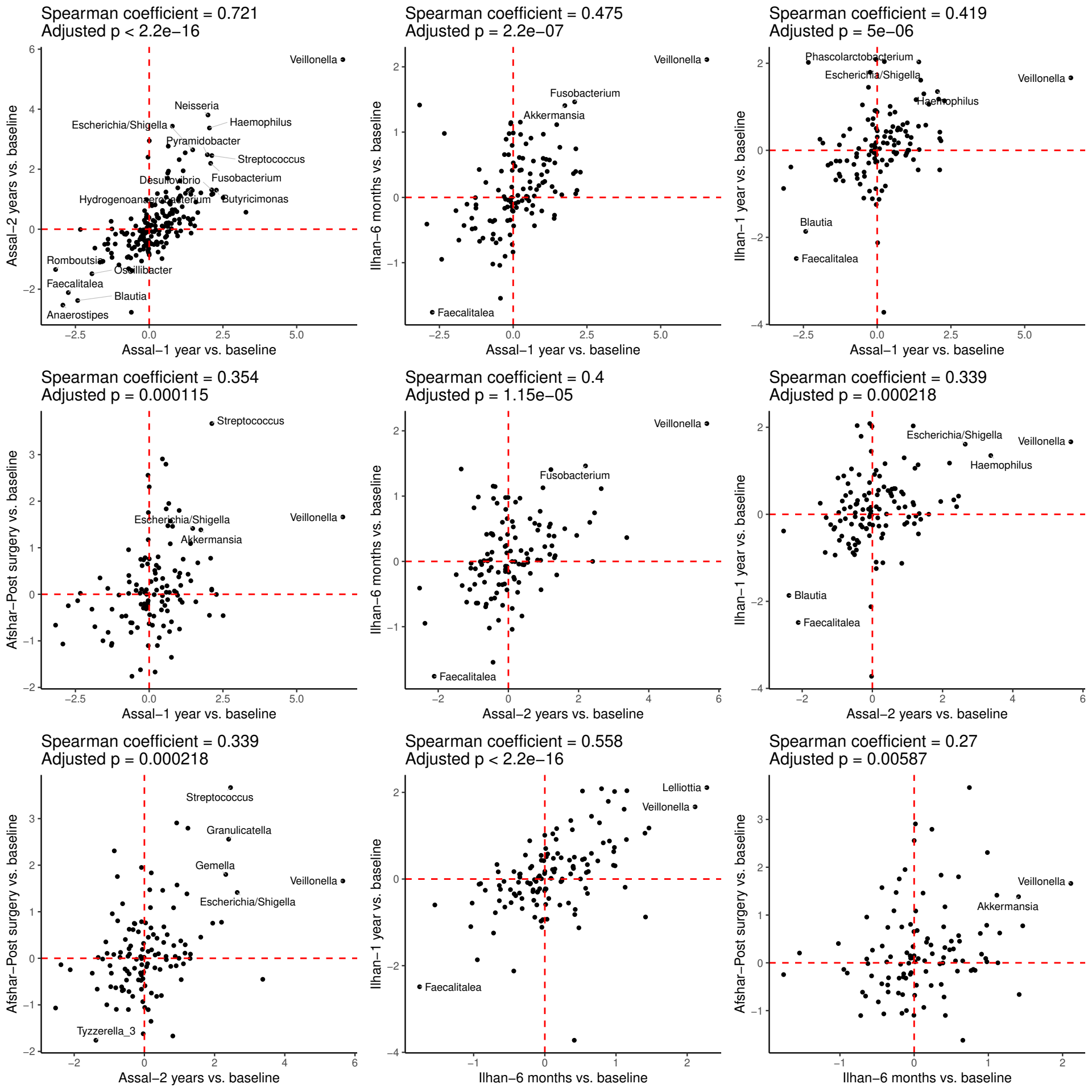
Heatmap of all pairwise Spearman r among the 4 original cohorts

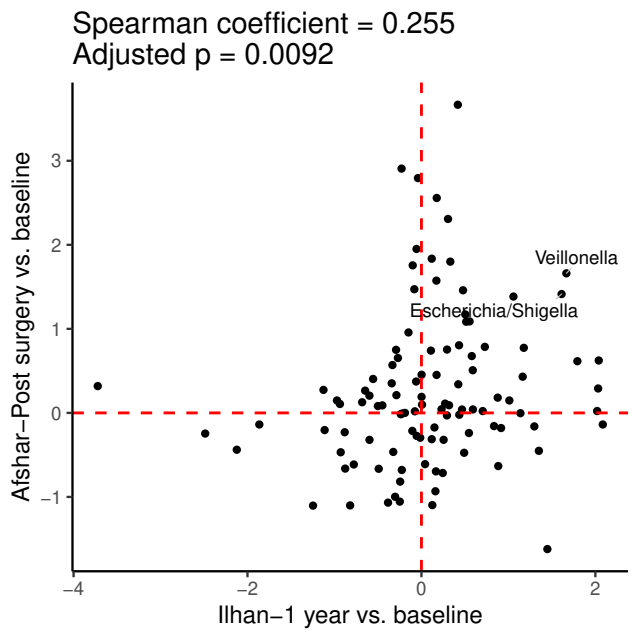
followed by the full set of pairwise scatter plots


Pairwise Spearman r (signed −log10 p) between study−timepoints
Cell text: r and BH−adjusted p. Grey = same study (excluded from headline r). Colour: 0.20 (blue) to 0.72 (red)
BS−p1M
BS−p6M
Assal−p3M
Assal−p1Y
Assal−p2Y
Ilhan−p6M
Ilhan−p1Y
Afshar−p6M
r=0.56 p=<0.001
r=0.66 p=<0.001
r=0.40 p=<0.001
r=0.50 p=<0.001
r=0.41 p=<0.001
r=0.20 p=0.031
r=0.49 p=<0.001
r=0.56 p=<0.001
r=0.43 p=<0.001
r=0.44 p=<0.001
r=0.43 p=<0.001
r=0.33 p=<0.001
r=0.26 p=0.005
r=0.43 p=<0.001
r=0.66 p=<0.001
r=0.43 p=<0.001
r=0.61 p=<0.001
r=0.65 p=<0.001
r=0.44 p=<0.001
r=0.36 p=<0.001
r=0.49 p=<0.001
r=0.40 p=<0.001
r=0.44 p=<0.001
r=0.61 p=<0.001
r=0.72 p=<0.001
r=0.48 p=<0.001
r=0.42 p=<0.001
r=0.35 p=<0.001
r=0.50 p=<0.001
r=0.43 p=<0.001
r=0.65 p=<0.001
r=0.72 p=<0.001
r=0.40 p=<0.001
r=0.34 p=<0.001
r=0.34 p=<0.001
r=0.41 p=<0.001
r=0.33 p=<0.001
r=0.44 p=<0.001
r=0.48 p=<0.001
r=0.40 p=<0.001
r=0.56 p=<0.001
r=0.27 p=0.006
r=0.20 p=0.031
r=0.26 p=0.005
r=0.36 p=<0.001
r=0.42 p=<0.001
r=0.34 p=<0.001
r=0.56 p=<0.001
r=0.25 p=0.009
r=0.49 p=<0.001
r=0.43 p=<0.001
r=0.49 p=<0.001
r=0.35 p=<0.001
r=0.34 p=<0.001
r=0.27 p=0.006
r=0.25 p=0.009
BS−p1M
BS−p6M
Assal−p3M
Assal−p1Y
Assal−p2Y
Ilhan−p6M
Ilhan−p1Y
Afshar−p6M


Spearman coefficient = 0.56
Adjusted p < 2.2e−16
BS−6 months vs. baseline
BS−1 month vs. baseline
Streptococcus
Enterococcus
Rothia
Veillonella
Actinomyces
Clostridium_sensu_stricto_1
Fusobacterium
Ruminiclostridium_5
Blautia
Spearman coefficient = 0.662
Adjusted p < 2.2e−16
Assal−3 months vs. baseline
BS−1 month vs. baseline
Veillonella
Atopobium
Gemella
Peptostreptococcus
Streptococcus
Escherichia/Shigella
Haemophilus
Oribacterium
Actinomyces
Rothia
Mogibacterium
Butyricimonas
Fusobacterium
Granulicatella
Parvimonas
Solobacterium
Romboutsia
Turicibacter
Anaerostipes
Agathobacter
Faecalibacterium
Spearman coefficient = 0.399
Adjusted p = 5e−06
Assal−1 year vs. baseline
BS−1 month vs. baseline
Veillonella
Butyricimonas
Mogibacterium
Streptococcus
Haemophilus
Fusobacterium
Escherichia/Shigella
Turicibacter
Blautia
Agathobacter
Anaerostipes
Romboutsia
Spearman coefficient = 0.497
Adjusted p = 9.84e−09
Assal−2 years vs. baseline
BS−1 month vs. baseline
Veillonella
Haemophilus
Escherichia/Shigella
Gemella
Granulicatella
Oribacterium
Streptococcus
Fusobacterium
Butyricimonas
Solobacterium
Coprococcus_3
Romboutsia
Blautia
Anaerostipes
Spearman coefficient = 0.415
Adjusted p = 6.78e−06
Ilhan−6 months vs. baseline
BS−1 month vs. baseline
Veillonella
Fusobacterium
Spearman coefficient = 0.2
Adjusted p = 0.0308
Ilhan−1 year vs. baseline
BS−1 month vs. baseline
Clostridium_sensu_stricto_1
Escherichia/Shigella
Veillonella
Haemophilus
Blautia
Spearman coefficient = 0.486
Adjusted p = 3.55e−07
Afshar−Post surgery vs. baseline
BS−1 month vs. baseline
Streptococcus
Actinomyces
Rothia
Enterococcus
Granulicatella
Gemella
Veillonella
Atopobium
Escherichia/Shigella
Faecalibacterium
Spearman coefficient = 0.428
Adjusted p = 9.88e−07
Assal−3 months vs. baseline
BS−6 months vs. baseline
Veillonella
Streptococcus
Akkermansia
Actinomyces
Rothia
Fusobacterium
Klebsiella
Spearman coefficient = 0.436
Adjusted p = 6.49e−07
Assal−1 year vs. baseline
BS−6 months vs. baseline
Veillonella
Fusobacterium
Streptococcus
Akkermansia
Blautia


Spearman coefficient = 0.434
Adjusted p = 7.04e−07
Assal−2 years vs. baseline
BS−6 months vs. baseline
Veillonella
Fusobacterium
Streptococcus
Blautia
Spearman coefficient = 0.329
Adjusted p = 0.000368
Ilhan−6 months vs. baseline
BS−6 months vs. baseline
Veillonella
Fusobacterium
Akkermansia
Spearman coefficient = 0.262
Adjusted p = 0.00502
Ilhan−1 year vs. baseline
BS−6 months vs. baseline
Clostridium_sensu_stricto_1
Veillonella
Blautia
Spearman coefficient = 0.427
Adjusted p = 7.24e−06
Afshar−Post surgery vs. baseline
BS−6 months vs. baseline
Streptococcus
Rothia
Actinomyces
Enterococcus
Veillonella
Ruminococcaceae_NK4A214_group
Akkermansia
Spearman coefficient = 0.608
Adjusted p < 2.2e−16
Assal−1 year vs. baseline
Assal−3 months vs. baseline
Veillonella
Butyricimonas
Desulfovibrio
Mogibacterium
Streptococcus
Haemophilus
Neisseria
Akkermansia
Escherichia/Shigella
Turicibacter
Agathobacter
Faecalitalea
Anaerostipes
Romboutsia
Spearman coefficient = 0.651
Adjusted p < 2.2e−16
Assal−2 years vs. baseline
Assal−3 months vs. baseline
Veillonella
Haemophilus
Neisseria
Actinobacillus
Escherichia/Shigella
Anaeroglobus
Granulicatella
Streptococcus
Solobacterium
Leptotrichia
Gemella
Selenomonas
Fusobacterium
Oribacterium
Desulfovibrio
Butyricimonas
Romboutsia
Tyzzerella_3
Faecalitalea
Anaerostipes
Succinivibrio
Spearman coefficient = 0.445
Adjusted p = 1.09e−06
Ilhan−6 months vs. baseline
Assal−3 months vs. baseline
Veillonella
Fusobacterium
Akkermansia
Faecalitalea
Spearman coefficient = 0.36
Adjusted p = 9.03e−05
Ilhan−1 year vs. baseline
Assal−3 months vs. baseline
Alistipes
Escherichia/Shigella
Veillonella
Haemophilus
Faecalitalea
Spearman coefficient = 0.487
Adjusted p = 1.01e−07
Afshar−Post surgery vs. baseline
Assal−3 months vs. baseline
Streptococcus
Actinomyces
Rothia
Granulicatella
Anaerotruncus
Gemella
Veillonella
Atopobium
Akkermansia
Escherichia/Shigella
Faecalibacterium
Tyzzerella_3


Spearman coefficient = 0.721
Adjusted p < 2.2e−16
Assal−2 years vs. baseline
Assal−1 year vs. baseline
Veillonella
Neisseria
Haemophilus
Escherichia/Shigella
Pyramidobacter
Streptococcus
Fusobacterium
Desulfovibrio
Hydrogenoanaerobacterium
Butyricimonas
Romboutsia
Oscillibacter
Faecalitalea
Blautia
Anaerostipes
Spearman coefficient = 0.475
Adjusted p = 2.2e−07
Ilhan−6 months vs. baseline
Assal−1 year vs. baseline
Veillonella
Fusobacterium
Akkermansia
Faecalitalea
Spearman coefficient = 0.419
Adjusted p = 5e−06
Ilhan−1 year vs. baseline
Assal−1 year vs. baseline
Phascolarctobacterium
Escherichia/Shigella
Veillonella
Haemophilus
Blautia
Faecalitalea
Spearman coefficient = 0.354
Adjusted p = 0.000115
Afshar−Post surgery vs. baseline
Assal−1 year vs. baseline
Streptococcus
Escherichia/Shigella
Veillonella
Akkermansia
Spearman coefficient = 0.4
Adjusted p = 1.15e−05
Ilhan−6 months vs. baseline
Assal−2 years vs. baseline
Veillonella
Fusobacterium
Faecalitalea
Spearman coefficient = 0.339
Adjusted p = 0.000218
Ilhan−1 year vs. baseline
Assal−2 years vs. baseline
Escherichia/Shigella
Veillonella
Haemophilus
Blautia
Faecalitalea
Spearman coefficient = 0.339
Adjusted p = 0.000218
Afshar−Post surgery vs. baseline
Assal−2 years vs. baseline
Streptococcus
Granulicatella
Gemella
Veillonella
Escherichia/Shigella
Tyzzerella_3
Spearman coefficient = 0.558
Adjusted p < 2.2e−16
Ilhan−1 year vs. baseline
Ilhan−6 months vs. baseline
Lelliottia
Veillonella
Faecalitalea
Spearman coefficient = 0.27
Adjusted p = 0.00587
Afshar−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Veillonella
Akkermansia


Spearman coefficient = 0.255
Adjusted p = 0.0092
Veillonella
Escherichia/Shigella
Afshar−Post surgery vs. baseline
Ilhan−1 year vs. baseline
3
2
1
0
−1
−4
−2
0
2

Phase 2: Extension to non–bariatric GI surgery

New cohorts: Cholecystectomy, IleocecalResection, Ileostomy, SDT

Scatter plots: each new cohort vs. each OG cohort only

(within–OG and within–new–cohort pairs omitted – already shown elsewhere)

Heatmap: all 8 cohorts together

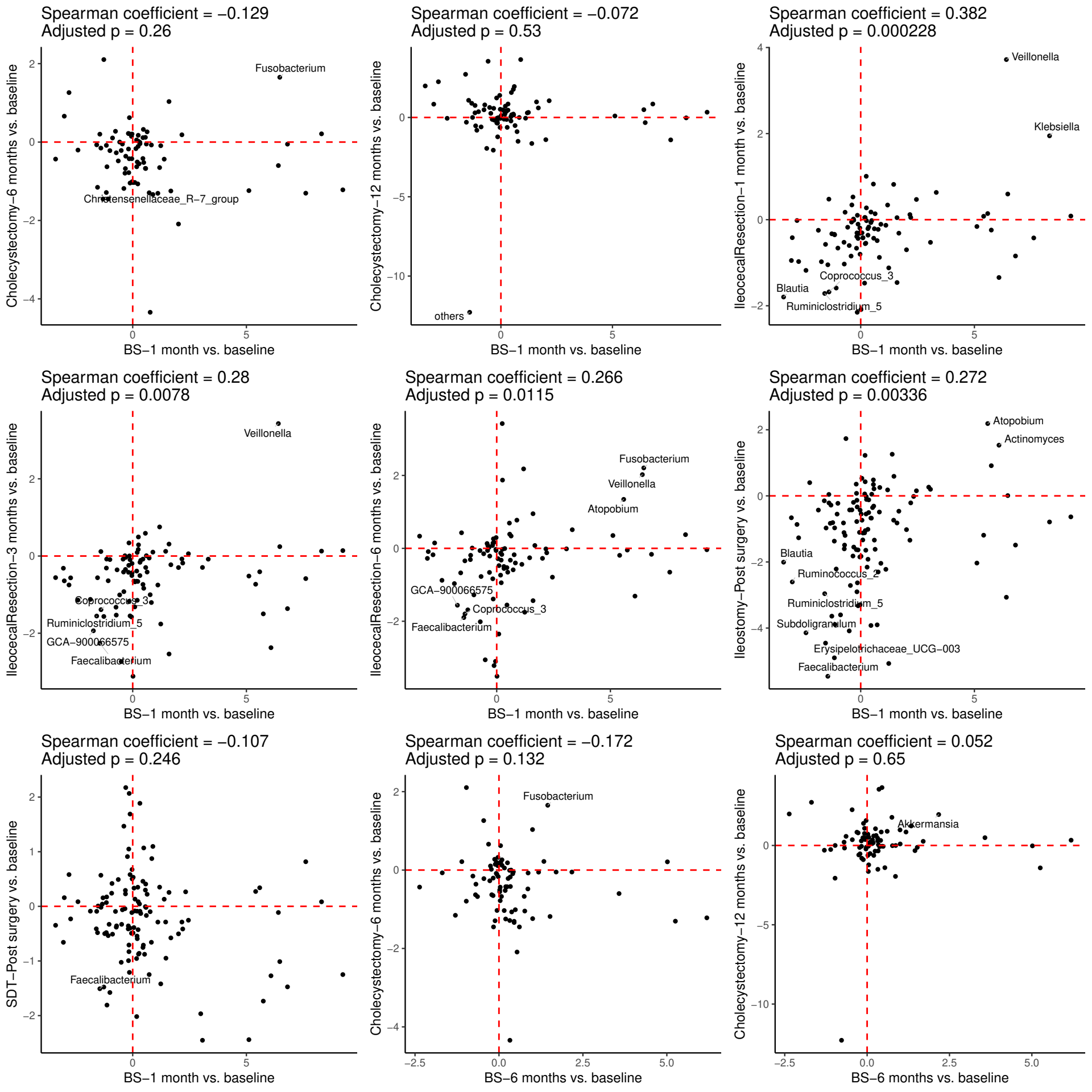
Phase 2: New GI−surgery cohorts vs. OG (RYGB) cohorts

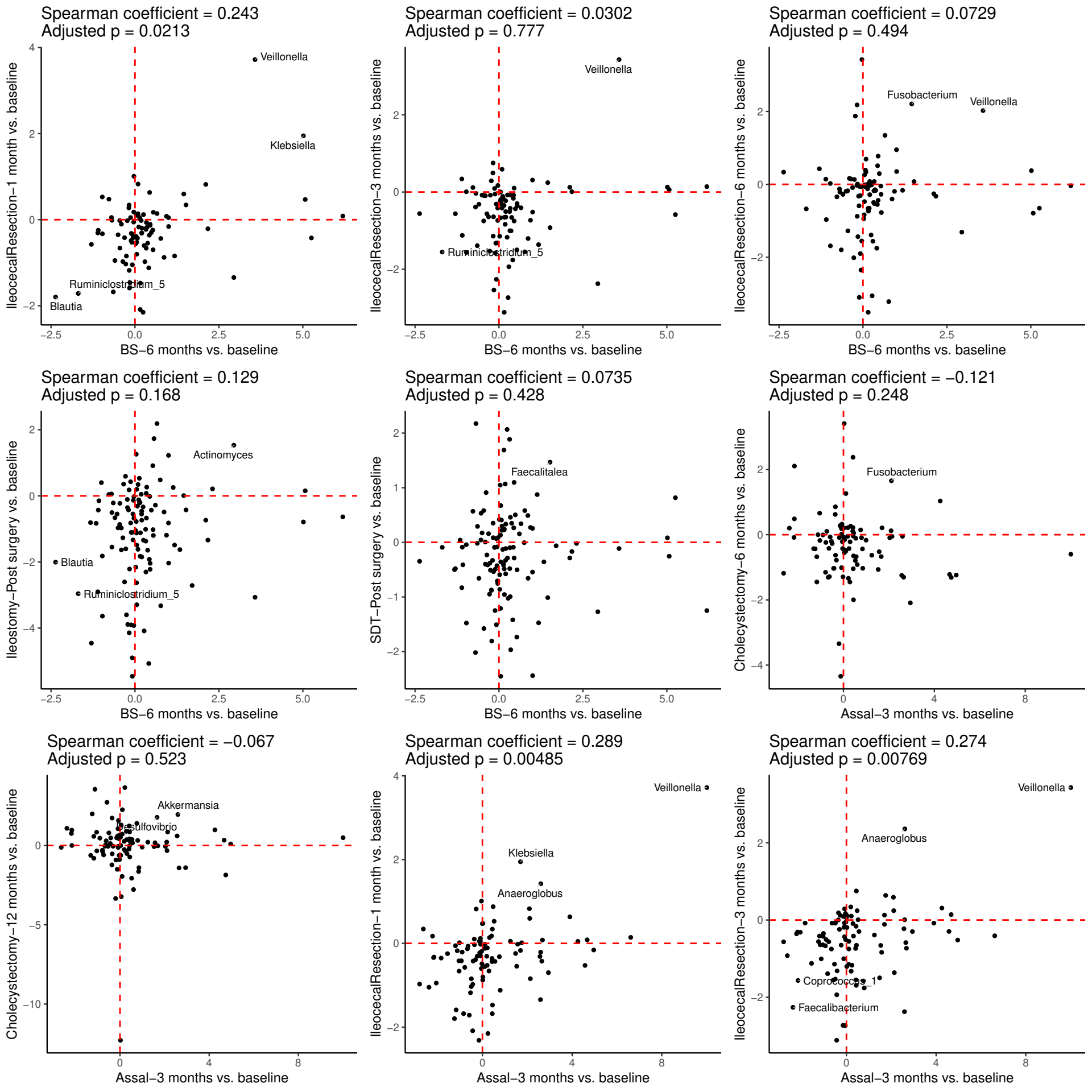
Group A: BS, Assal, Ilhan, Afshar

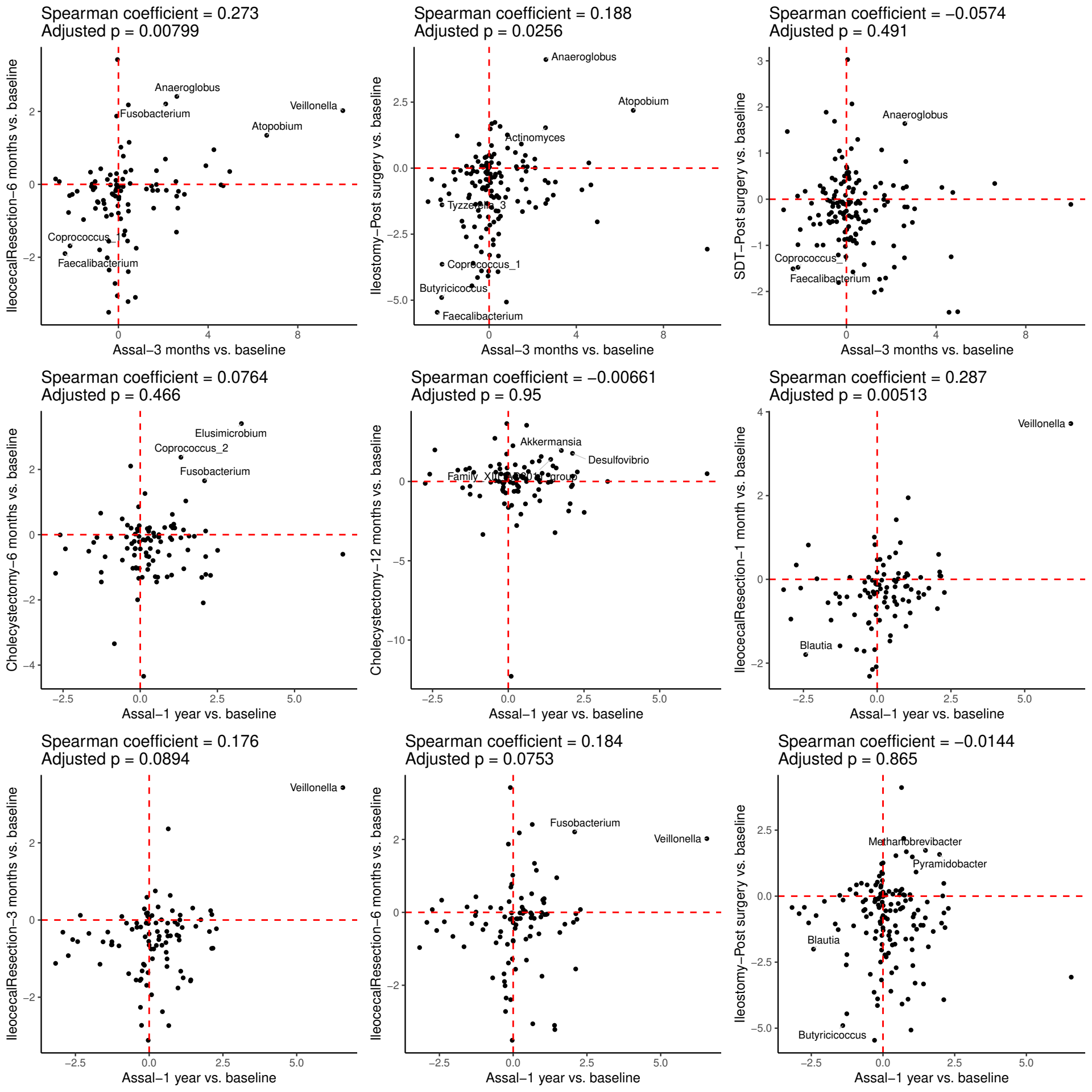
Group B: Cholecystectomy, IleocecalResection, Ileostomy, SDT

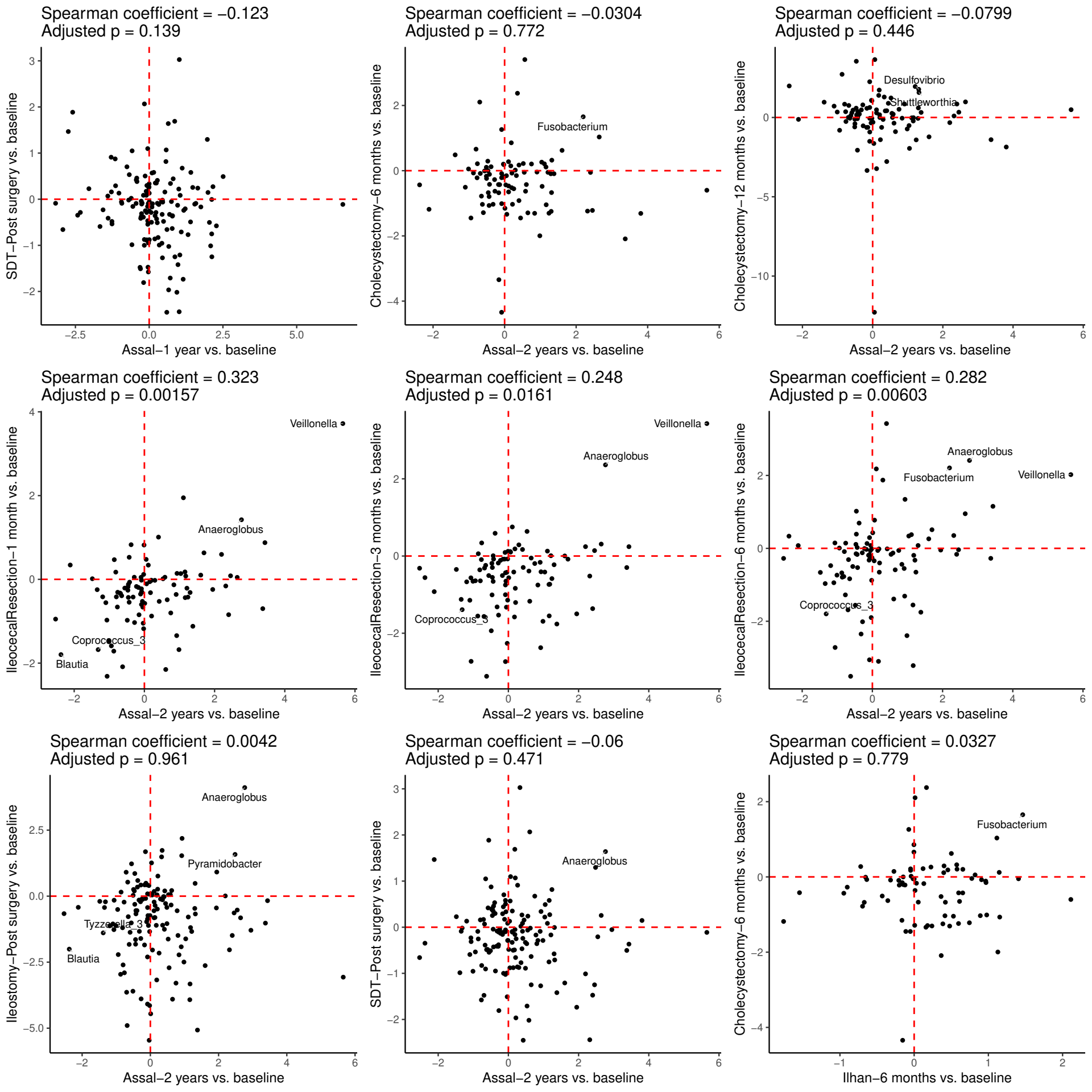
56 cross−group pairs (within−group pairs excluded)

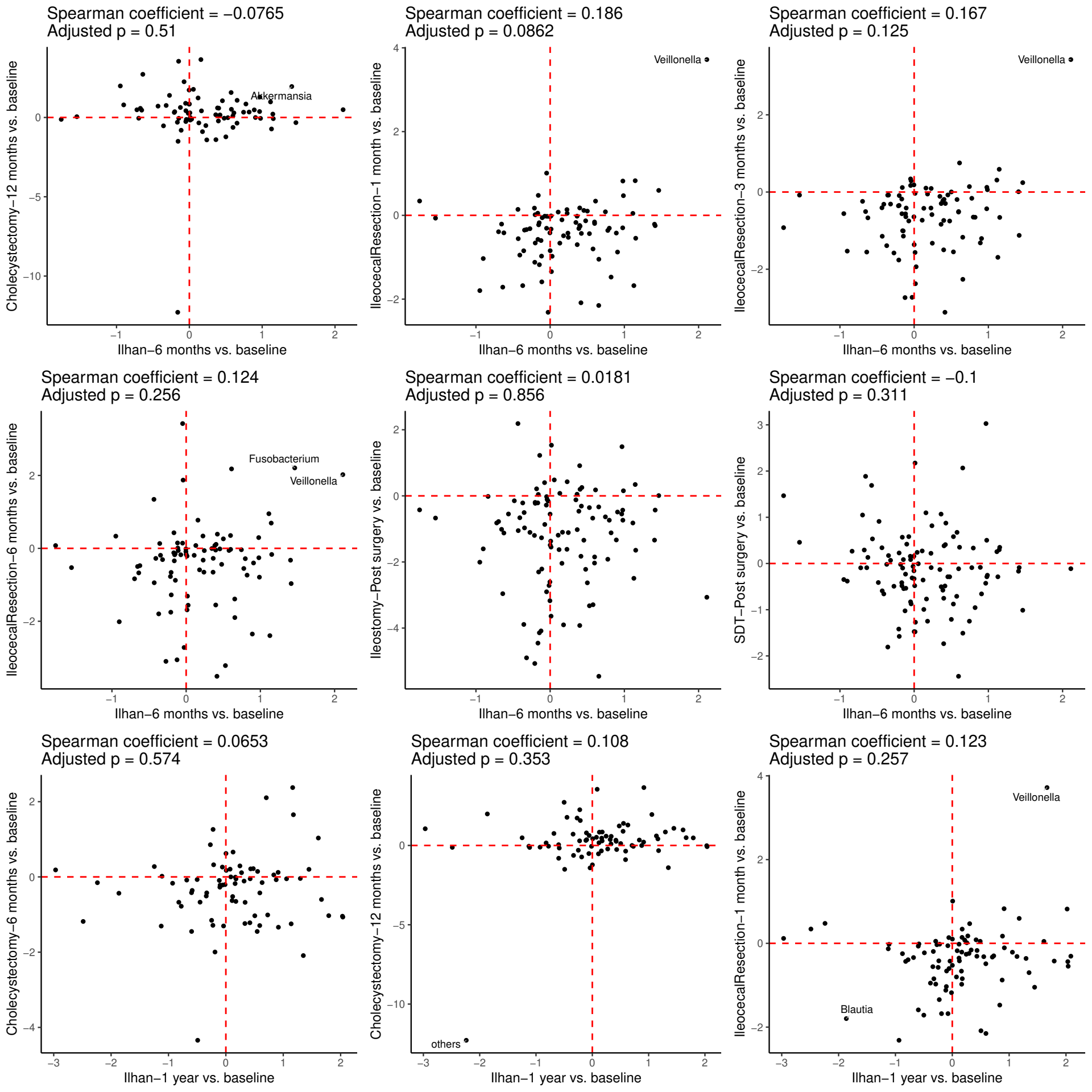
Each point: one taxon's signed −log10(p). r = Spearman correlation, p = BH−adjusted.

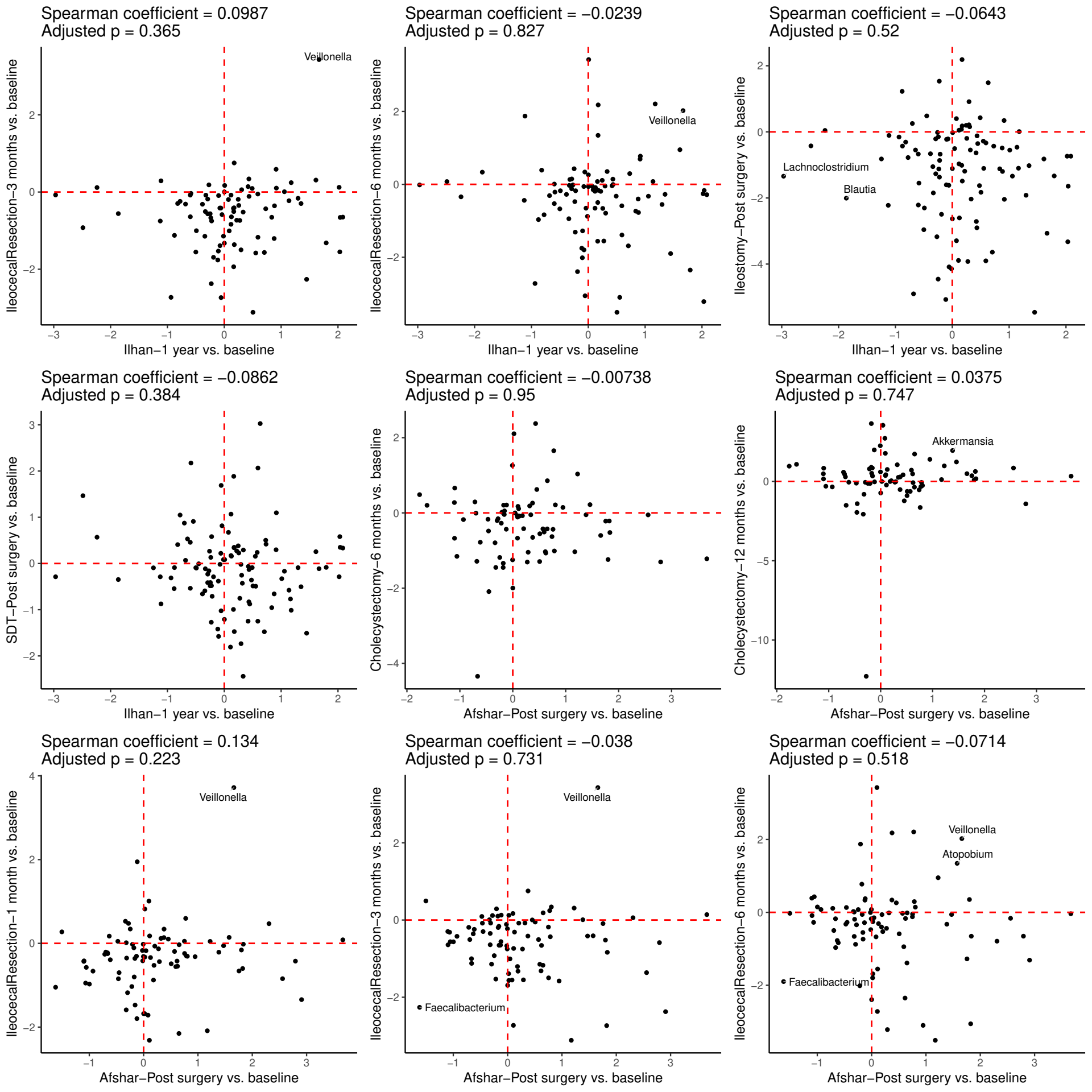

Spearman coefficient = −0.129
Adjusted p = 0.26
Fusobacterium
Christensenellaceae_R−7_group
Cholecystectomy−6 months vs. baseline
BS−1 month vs. baseline
Spearman coefficient = −0.072
Adjusted p = 0.53
others
Cholecystectomy−12 months vs. baseline
BS−1 month vs. baseline
Spearman coefficient = 0.382
Adjusted p = 0.000228
Veillonella
Klebsiella
Coprococcus_3
Blautia
Ruminiclostridium_5
IleocecalResection−1 month vs. baseline
BS−1 month vs. baseline
Spearman coefficient = 0.28
Adjusted p = 0.0078
Veillonella
Coprococcus_3
Ruminiclostridium_5
GCA−900066575
Faecalibacterium
IleocecalResection−3 months vs. baseline
BS−1 month vs. baseline
Spearman coefficient = 0.266
Adjusted p = 0.0115
Fusobacterium
Veillonella
Atopobium
GCA−900066575
Coprococcus_3
Faecalibacterium
IleocecalResection−6 months vs. baseline
BS−1 month vs. baseline
Spearman coefficient = 0.272
Adjusted p = 0.00336
Atopobium
Actinomyces
Blautia
Ruminococcus_2
Ruminiclostridium_5
Subdoligranulum
Erysipelotrichaceae_UCG−003
Faecalibacterium
Ileostomy−Post surgery vs. baseline
BS−1 month vs. baseline
Spearman coefficient = −0.107
Adjusted p = 0.246
Faecalibacterium
SDT−Post surgery vs. baseline
BS−1 month vs. baseline
Spearman coefficient = −0.172
Adjusted p = 0.132
Fusobacterium
Cholecystectomy−6 months vs. baseline
BS−6 months vs. baseline
Spearman coefficient = 0.052
Adjusted p = 0.65
Akkermansia
Cholecystectomy−12 months vs. baseline
BS−6 months vs. baseline

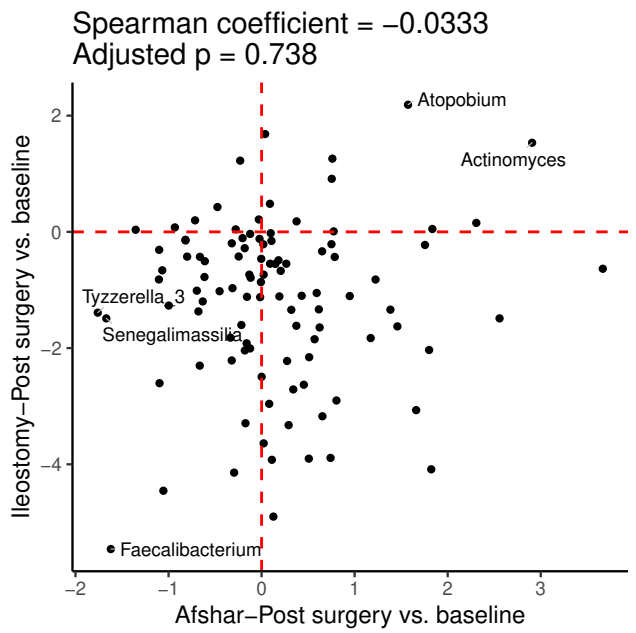

Spearman coefficient = 0.243
Adjusted p = 0.0213
Veillonella
Klebsiella
Ruminiclostridium_5
Blautia
IleocecalResection−1 month vs. baseline
BS−6 months vs. baseline
Spearman coefficient = 0.0302
Adjusted p = 0.777
Veillonella
Ruminiclostridium_5
IleocecalResection−3 months vs. baseline
BS−6 months vs. baseline
Spearman coefficient = 0.0729
Adjusted p = 0.494
Fusobacterium
Veillonella
IleocecalResection−6 months vs. baseline
BS−6 months vs. baseline
Spearman coefficient = 0.129
Adjusted p = 0.168
Actinomyces
Blautia
Ruminiclostridium_5
Ileostomy−Post surgery vs. baseline
BS−6 months vs. baseline
Spearman coefficient = 0.0735
Adjusted p = 0.428
Faecalitalea
SDT−Post surgery vs. baseline
BS−6 months vs. baseline
Spearman coefficient = −0.121
Adjusted p = 0.248
Fusobacterium
Cholecystectomy−6 months vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = −0.067
Adjusted p = 0.523
Akkermansia
Desulfovibrio
Cholecystectomy−12 months vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.289
Adjusted p = 0.00485
Veillonella
Klebsiella
Anaeroglobus
IleocecalResection−1 month vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.274
Adjusted p = 0.00769
Veillonella
Anaeroglobus
Coprococcus_1
Faecalibacterium
IleocecalResection−3 months vs. baseline
Assal−3 months vs. baseline

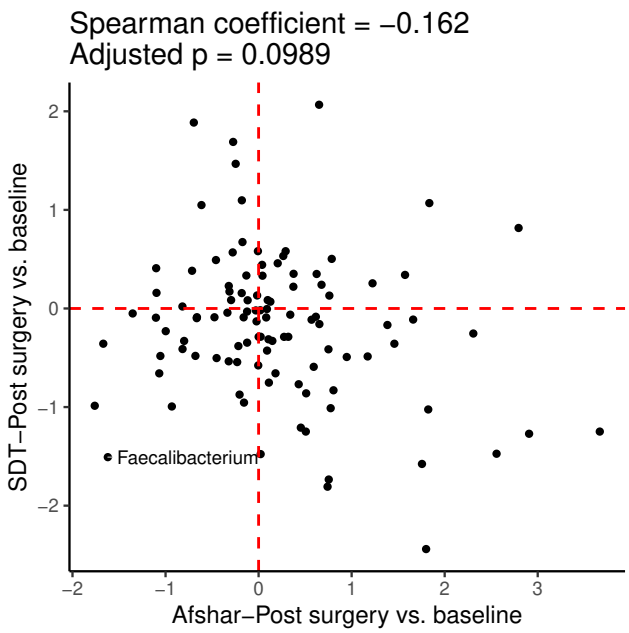

Spearman coefficient = 0.273
Adjusted p = 0.00799
Anaeroglobus
Veillonella
Fusobacterium
Atopobium
Coprococcus_1
Faecalibacterium
IleocecalResection−6 months vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.188
Adjusted p = 0.0256
Anaeroglobus
Atopobium
Actinomyces
Tyzzerella_3
Coprococcus_1
Butyricicoccus
Faecalibacterium
Ileostomy−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = −0.0574
Adjusted p = 0.491
Anaeroglobus
Coprococcus_1
Faecalibacterium
SDT−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.0764
Adjusted p = 0.466
Elusimicrobium
Coprococcus_2
Fusobacterium
Cholecystectomy−6 months vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = −0.00661
Adjusted p = 0.95
Akkermansia
Desulfovibrio
Cholecystectomy−12 months vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.287
Adjusted p = 0.00513
Veillonella
Blautia
IleocecalResection−1 month vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.176
Adjusted p = 0.0894
Veillonella
IleocecalResection−3 months vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.184
Adjusted p = 0.0753
Fusobacterium
Veillonella
IleocecalResection−6 months vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = −0.0144
Adjusted p = 0.865
Methanobrevibacter
Pyramidobacter
Blautia
Butyricicoccus
Ileostomy−Post surgery vs. baseline
Assal−1 year vs. baseline


Spearman coefficient = −0.123
Adjusted p = 0.139
SDT−Post surgery vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = −0.0304
Adjusted p = 0.772
Cholecystectomy−6 months vs. baseline
Fusobacterium
Assal−2 years vs. baseline
Spearman coefficient = −0.0799
Adjusted p = 0.446
Cholecystectomy−12 months vs. baseline
Desulfovibrio
Shuttleworthia
Assal−2 years vs. baseline
Spearman coefficient = 0.323
Adjusted p = 0.00157
IleocecalResection−1 month vs. baseline
Veillonella
Anaeroglobus
Coprococcus_3
Blautia
Assal−2 years vs. baseline
Spearman coefficient = 0.248
Adjusted p = 0.0161
IleocecalResection−3 months vs. baseline
Veillonella
Anaeroglobus
Coprococcus_3
Assal−2 years vs. baseline
Spearman coefficient = 0.282
Adjusted p = 0.00603
IleocecalResection−6 months vs. baseline
Anaeroglobus
Fusobacterium
Veillonella
Coprococcus_3
Assal−2 years vs. baseline
Spearman coefficient = 0.0042
Adjusted p = 0.961
Ileostomy−Post surgery vs. baseline
Anaeroglobus
Pyramidobacter
Tyzzerella_3
Blautia
Assal−2 years vs. baseline
Spearman coefficient = −0.06
Adjusted p = 0.471
SDT−Post surgery vs. baseline
Anaeroglobus
Assal−2 years vs. baseline
Spearman coefficient = 0.0327
Adjusted p = 0.779
Cholecystectomy−6 months vs. baseline
Fusobacterium
Ilhan−6 months vs. baseline


Spearman coefficient = −0.0765
Adjusted p = 0.51
Akkermansia
Cholecystectomy−12 months vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.186
Adjusted p = 0.0862
Veillonella
IleocecalResection−1 month vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.167
Adjusted p = 0.125
Veillonella
IleocecalResection−3 months vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.124
Adjusted p = 0.256
Fusobacterium
Veillonella
IleocecalResection−6 months vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.0181
Adjusted p = 0.856
Ileostomy−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = −0.1
Adjusted p = 0.311
SDT−Post surgery vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.0653
Adjusted p = 0.574
Cholecystectomy−6 months vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.108
Adjusted p = 0.353
others
Cholecystectomy−12 months vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.123
Adjusted p = 0.257
Veillonella
Blautia
IleocecalResection−1 month vs. baseline
Ilhan−1 year vs. baseline

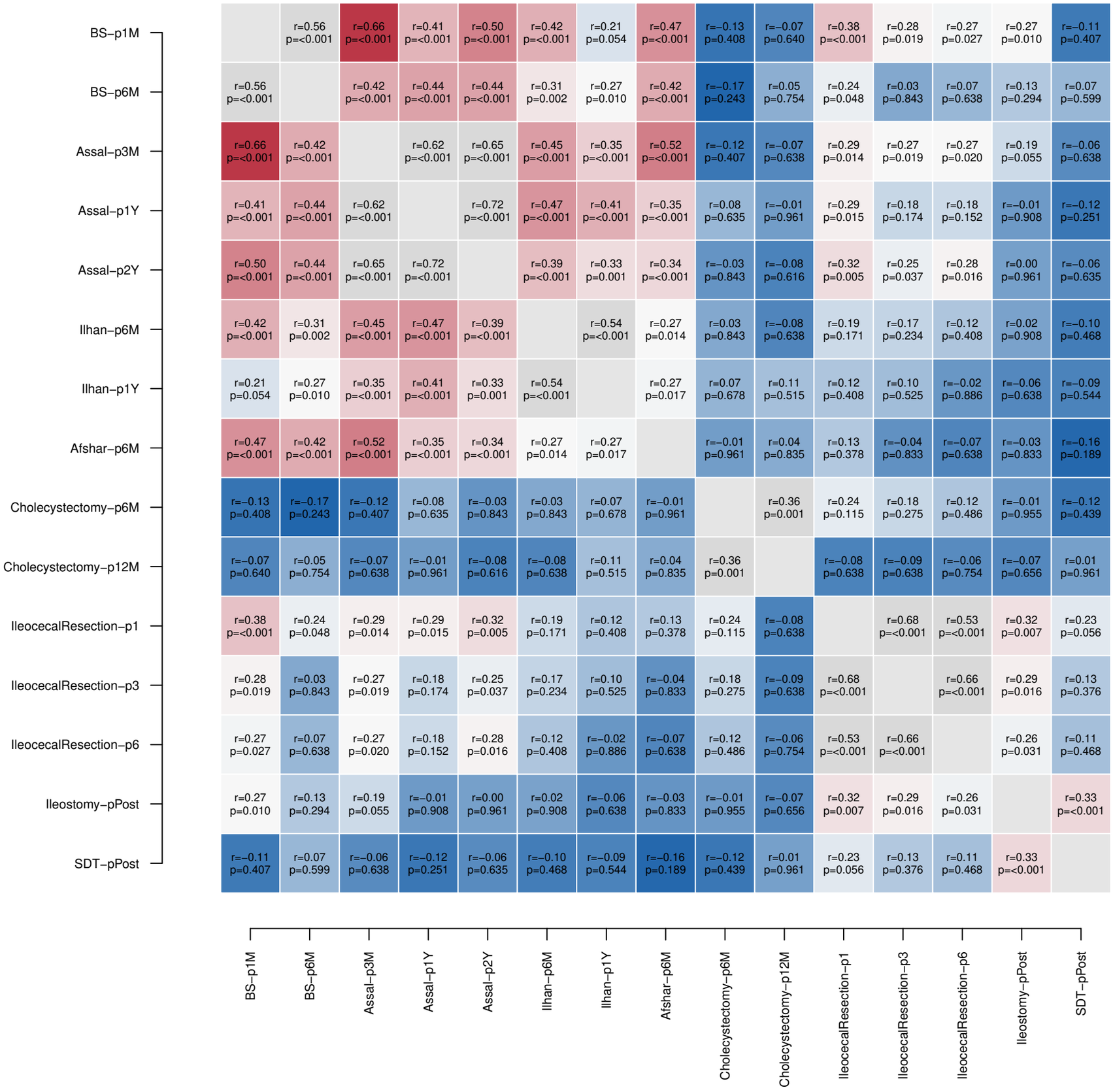

Spearman coefficient = 0.0987
Adjusted p = 0.365
Veillonella
IleocecalResection−3 months vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = −0.0239
Adjusted p = 0.827
Veillonella
IleocecalResection−6 months vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = −0.0643
Adjusted p = 0.52
Lachnoclostridium
Blautia
Ileostomy−Post surgery vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = −0.0862
Adjusted p = 0.384
SDT−Post surgery vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = −0.00738
Adjusted p = 0.95
Cholecystectomy−6 months vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = 0.0375
Adjusted p = 0.747
Akkermansia
Cholecystectomy−12 months vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = 0.134
Adjusted p = 0.223
Veillonella
IleocecalResection−1 month vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = −0.038
Adjusted p = 0.731
Veillonella
Faecalibacterium
IleocecalResection−3 months vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = −0.0714
Adjusted p = 0.518
Veillonella
Atopobium
Faecalibacterium
IleocecalResection−6 months vs. baseline
Afshar−Post surgery vs. baseline


Spearman coefficient = −0.0333
Adjusted p = 0.738
Atopobium
Actinomyces
Tyzzerella_3
Senegalimassilia
Faecalibacterium
Ileostomy−Post surgery vs. baseline
Afshar−Post surgery vs. baseline


Spearman coefficient = −0.162
Adjusted p = 0.0989
Faecalibacterium
SDT−Post surgery vs. baseline
Afshar−Post surgery vs. baseline

Pairwise Spearman r (signed −log10 p) between study−timepoints

Cell text: r and BH−adjusted p. Grey = same study (excluded from headline r). Colour: −0.17 (blue) to 0.72 (red)



Phase 3a: Positive control – non–RYGB bariatric

New cohorts: Chaudhari_SG, Lazaro_SG (both sleeve gastrectomy)

Scatter plots: each cohort vs. each OG cohort only

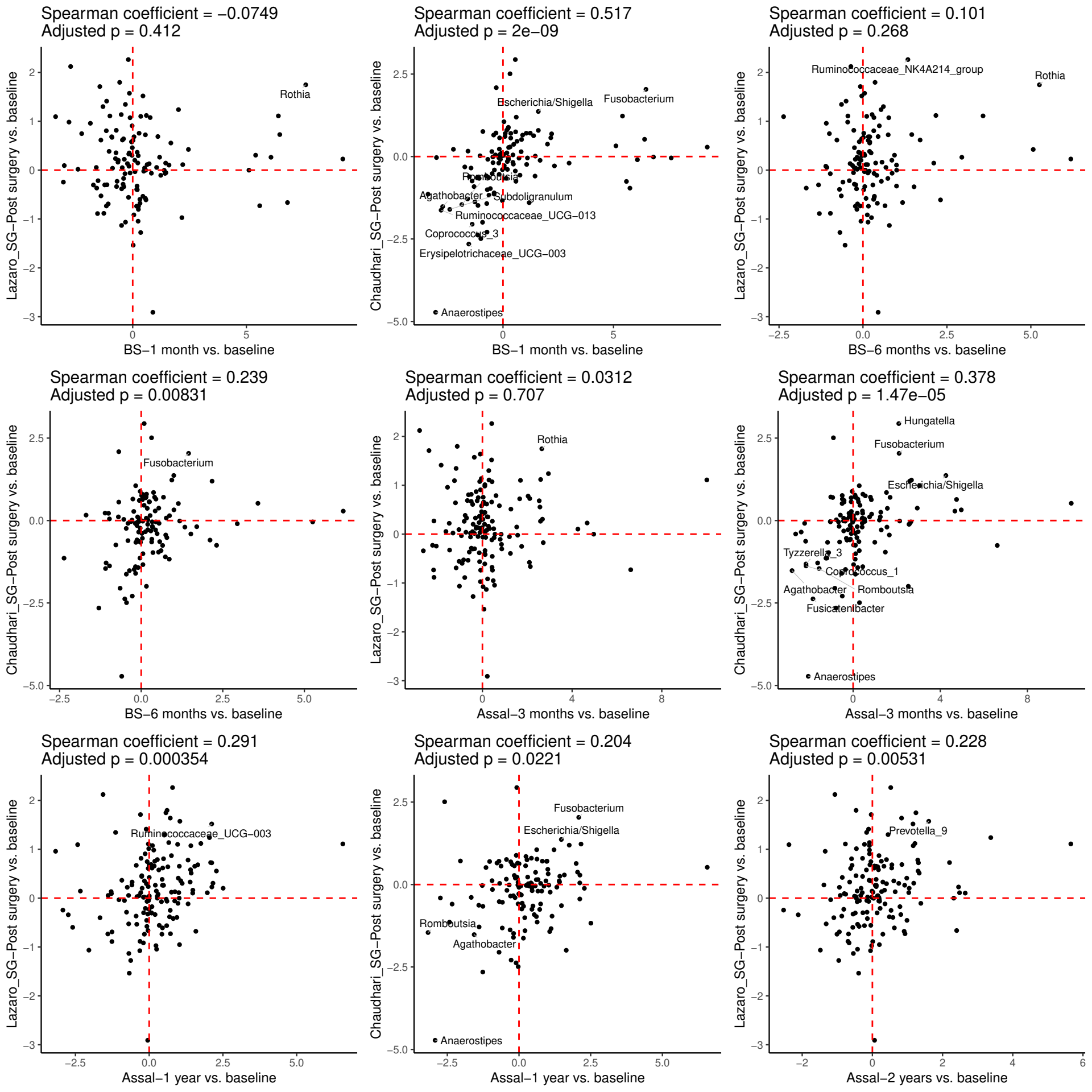
Phase 3a: Non−RYGB bariatric (sleeve gastrectomy) cohorts vs. OG

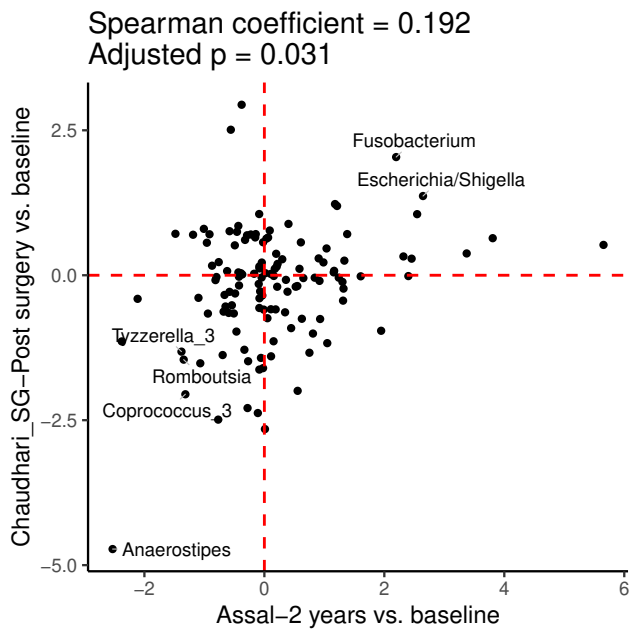
Group A: BS, Assal, Ilhan, Afshar

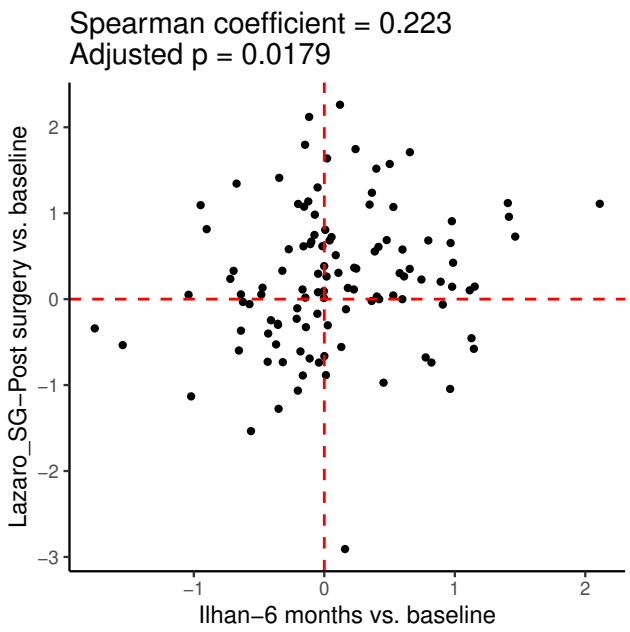
Group B: Chaudhari_SG, Lazaro_SG

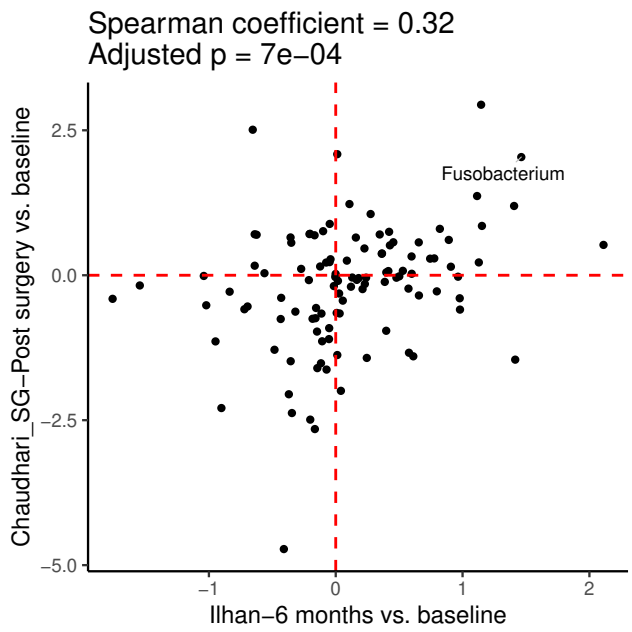
16 cross−group pairs (within−group pairs excluded)

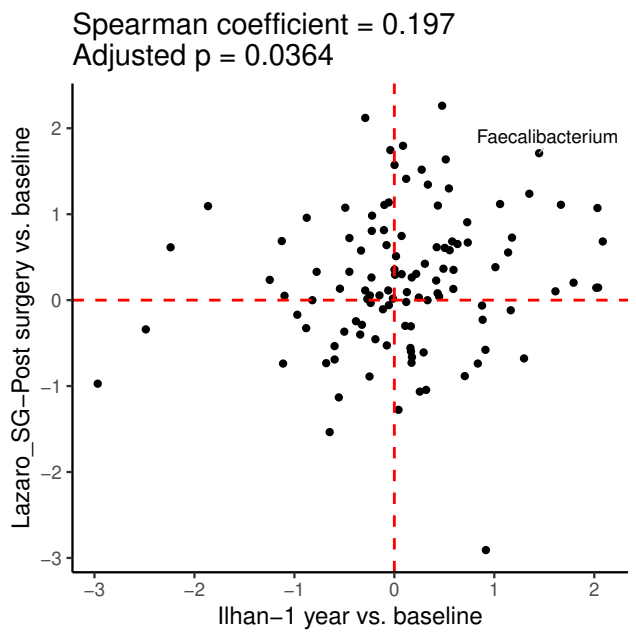
Each point: one taxon's signed −log10(p). r = Spearman correlation, p = BH−adjusted.

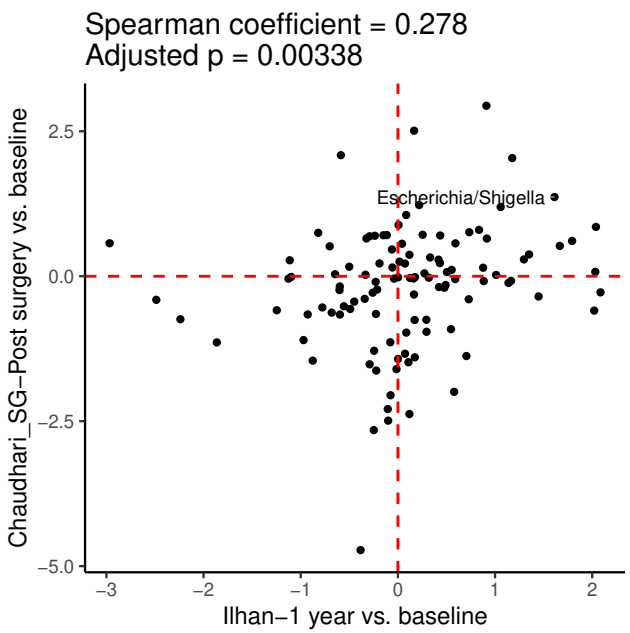

Spearman coefficient = −0.0749
Adjusted p = 0.412
Rothia
Lazaro_SG−Post surgery vs. baseline
BS−1 month vs. baseline
Spearman coefficient = 0.517
Adjusted p = 2e−09
Escherichia/Shigella
Fusobacterium
Romboutsia
Agathobacter
Subdoligranulum
Ruminococcaceae_UCG−013
Coprococcus_3
Erysipelotrichaceae_UCG−003
Anaerostipes
Chaudhari_SG−Post surgery vs. baseline
BS−1 month vs. baseline
Spearman coefficient = 0.101
Adjusted p = 0.268
Ruminococcaceae_NK4A214_group
Rothia
Lazaro_SG−Post surgery vs. baseline
BS−6 months vs. baseline
Spearman coefficient = 0.239
Adjusted p = 0.00831
Fusobacterium
Chaudhari_SG−Post surgery vs. baseline
BS−6 months vs. baseline
Spearman coefficient = 0.0312
Adjusted p = 0.707
Rothia
Lazaro_SG−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.378
Adjusted p = 1.47e−05
Hungatella
Fusobacterium
Escherichia/Shigella
Tyzzerella_3
Coprococcus_1
Agathobacter
Romboutsia
Fusicatenibacter
Anaerostipes
Chaudhari_SG−Post surgery vs. baseline
Assal−3 months vs. baseline
Spearman coefficient = 0.291
Adjusted p = 0.000354
Ruminococcaceae_UCG−003
Lazaro_SG−Post surgery vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.204
Adjusted p = 0.0221
Fusobacterium
Escherichia/Shigella
Romboutsia
Agathobacter
Anaerostipes
Chaudhari_SG−Post surgery vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.228
Adjusted p = 0.00531
Prevotella_9
Lazaro_SG−Post surgery vs. baseline
Assal−2 years vs. baseline

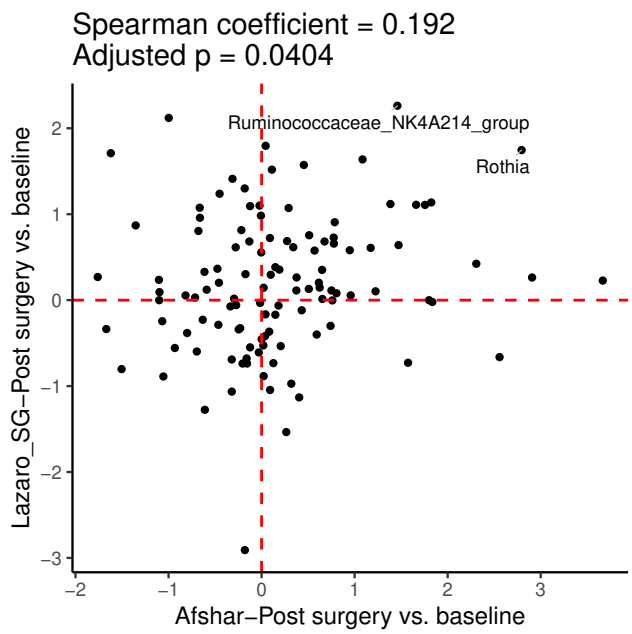

Spearman coefficient = 0.192
Adjusted p = 0.031
Fusobacterium
Escherichia/Shigella
Tyzzerella_3
Romboutsia
Coprococcus_3
Anaerostipes
Chaudhari_SG−Post surgery vs. baseline
Assal−2 years vs. baseline

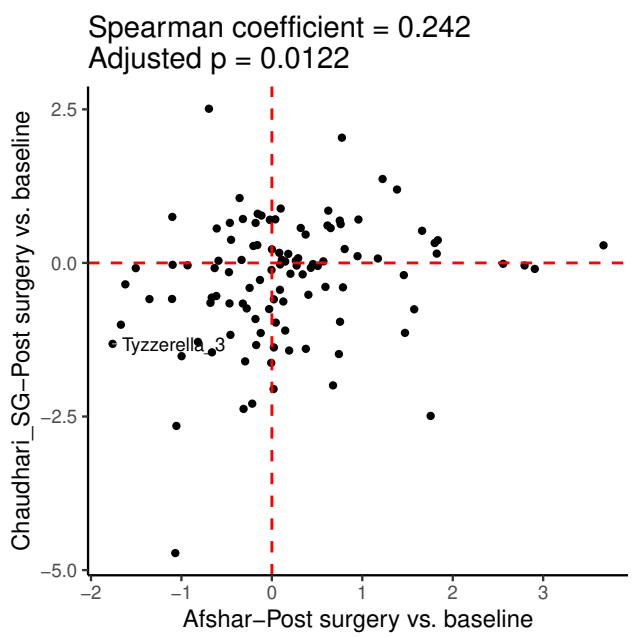

Spearman coefficient = 0.223
Adjusted p = 0.0179
Lazaro_SG−Post surgery vs. baseline
Ilhan−6 months vs. baseline


Spearman coefficient = 0.32
Adjusted p = 7e−04
Fusobacterium
Chaudhari_SG−Post surgery vs. baseline
Ilhan−6 months vs. baseline


Spearman coefficient = 0.197
Adjusted p = 0.0364
Faecalibacterium
Lazaro_SG−Post surgery vs. baseline
Ilhan−1 year vs. baseline


Spearman coefficient = 0.278
Adjusted p = 0.00338
Escherichia/Shigella
Chaudhari_SG−Post surgery vs. baseline
Ilhan−1 year vs. baseline


Spearman coefficient = 0.192
Adjusted p = 0.0404
Ruminococcaceae_NK4A214_group
Rothia
Lazaro_SG−Post surgery vs. baseline
Afshar−Post surgery vs. baseline


Spearman coefficient = 0.242
Adjusted p = 0.0122
Tyzzerella_3
Chaudhari_SG−Post surgery vs. baseline
Afshar−Post surgery vs. baseline

Phase 3b: Positive control – RYGB–specific

New cohorts: RYGB_SaoPaulo, Lazaro_GB (both RYGB)

Scatter plots: each cohort vs. each OG cohort only

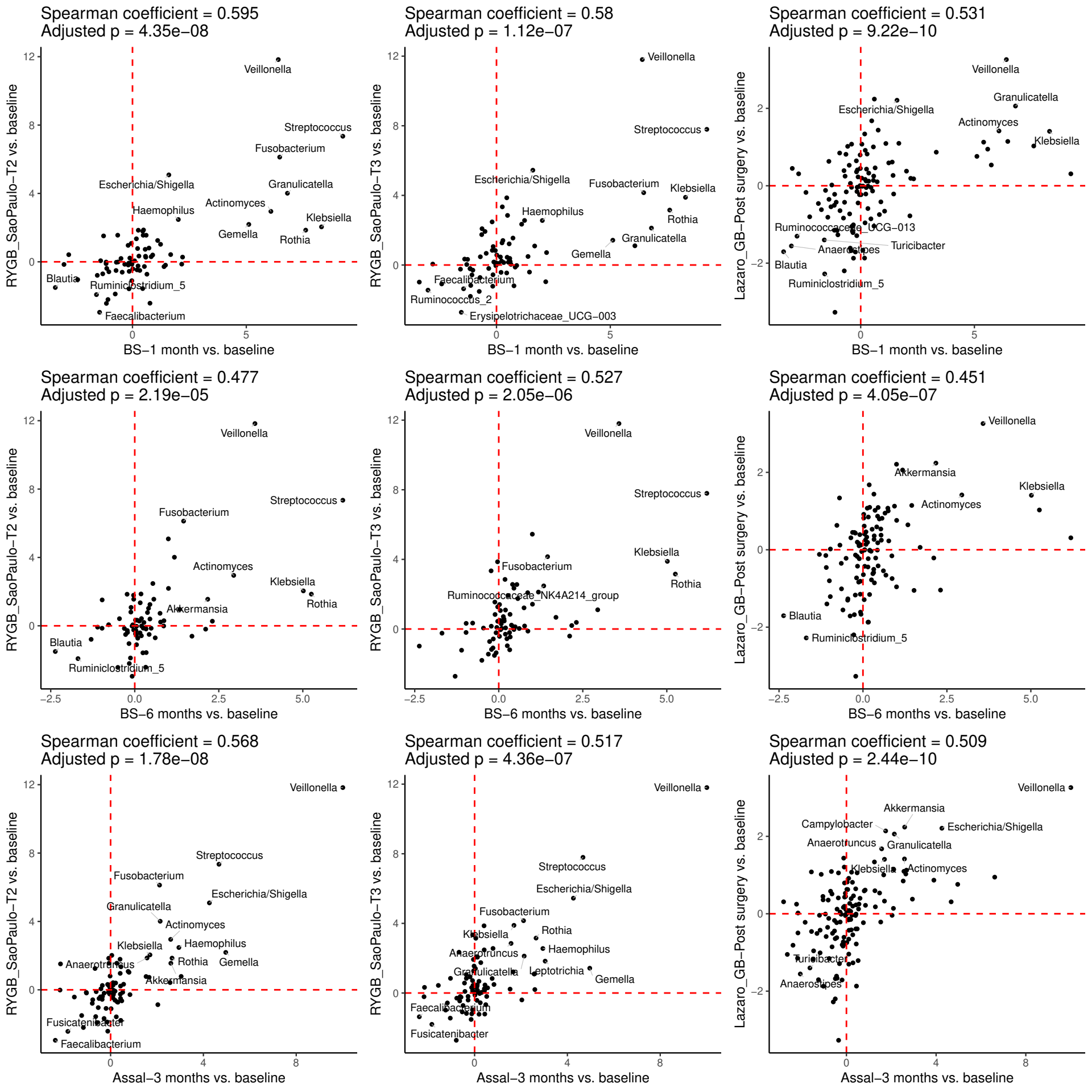
Phase 3b: RYGB−specific cohorts vs. OG

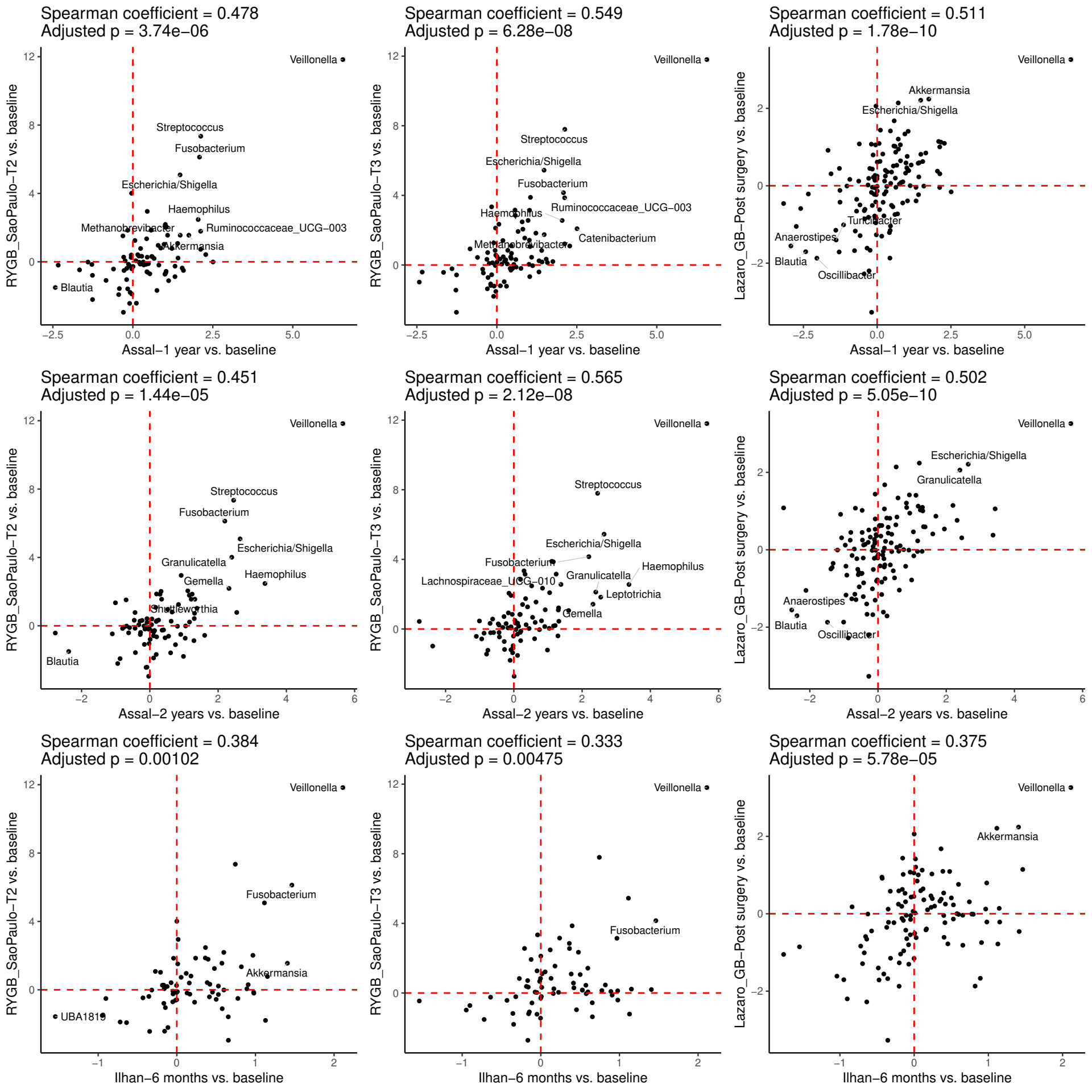
Group A: BS, Assal, Ilhan, Afshar

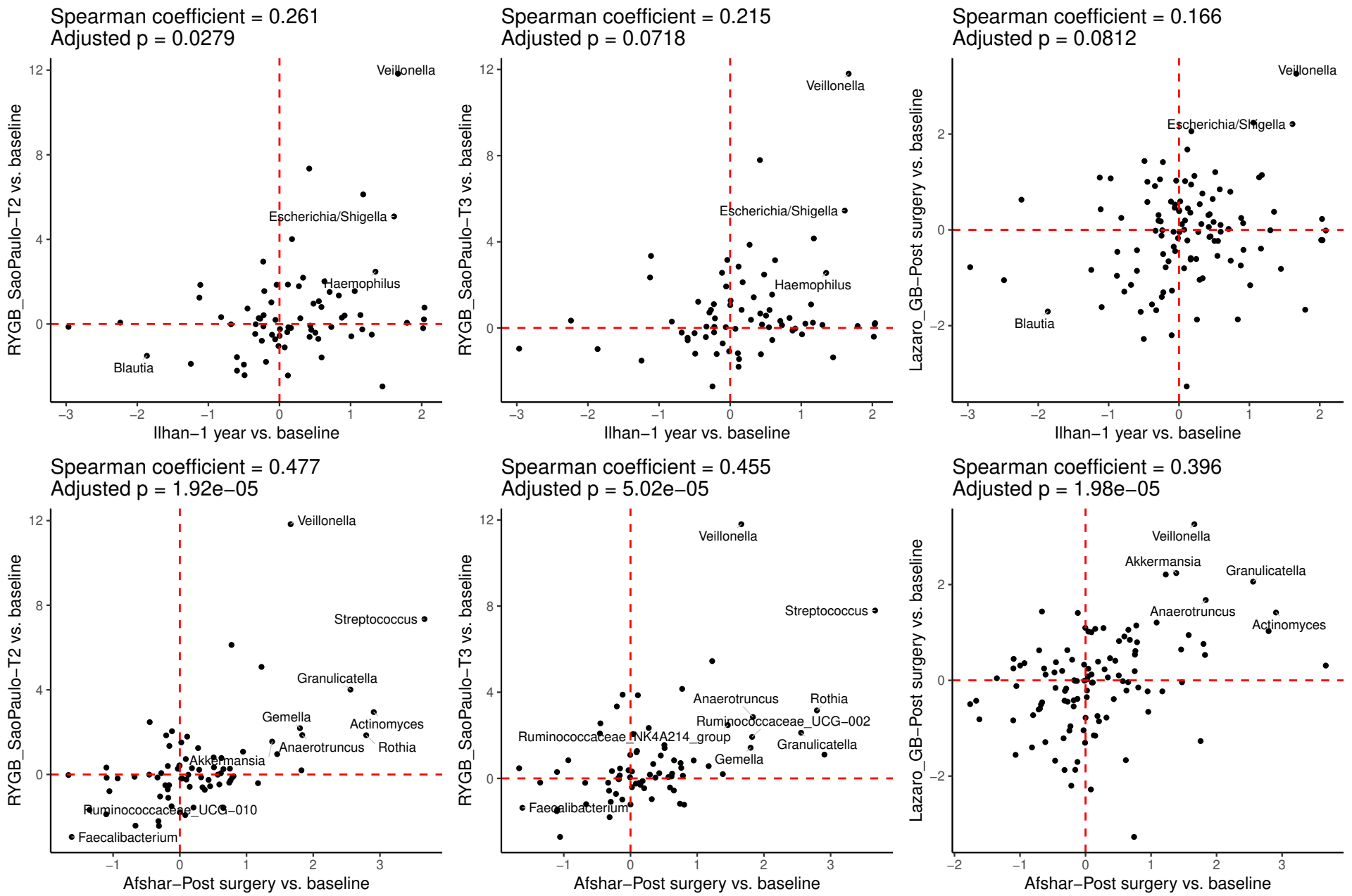
Group B: RYGB_SaoPaulo, Lazaro_GB

24 cross−group pairs (within−group pairs excluded)

Each point: one taxon's signed −log10(p). r = Spearman correlation, p = BH−adjusted.


Spearman coefficient = 0.595
Adjusted p = 4.35e−08
RYGB_SaoPaulo−T2 vs. baseline
BS−1 month vs. baseline
Veillonella
Streptococcus
Fusobacterium
Escherichia/Shigella
Granulicatella
Actinomyces
Haemophilus
Klebsiella
Gemella
Rothia
Blautia
Ruminiclostridium_5
Faecalibacterium
Spearman coefficient = 0.58
Adjusted p = 1.12e−07
RYGB_SaoPaulo−T3 vs. baseline
BS−1 month vs. baseline
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Klebsiella
Haemophilus
Rothia
Granulicatella
Gemella
Faecalibacterium
Ruminococcus_2
Erysipelotrichaceae_UCG−003
Spearman coefficient = 0.531
Adjusted p = 9.22e−10
Lazaro_GB−Post surgery vs. baseline
BS−1 month vs. baseline
Veillonella
Granulicatella
Escherichia/Shigella
Actinomyces
Klebsiella
Ruminococcaceae_UCG−013
Turicibacter
Anaerostipes
Blautia
Ruminiclostridium_5
Spearman coefficient = 0.477
Adjusted p = 2.19e−05
RYGB_SaoPaulo−T2 vs. baseline
BS−6 months vs. baseline
Veillonella
Streptococcus
Fusobacterium
Actinomyces
Klebsiella
Rothia
Akkermansia
Blautia
Ruminiclostridium_5
Spearman coefficient = 0.527
Adjusted p = 2.05e−06
RYGB_SaoPaulo−T3 vs. baseline
BS−6 months vs. baseline
Veillonella
Streptococcus
Klebsiella
Fusobacterium
Rothia
Ruminococcaceae_NK4A214_group
Spearman coefficient = 0.451
Adjusted p = 4.05e−07
Lazaro_GB−Post surgery vs. baseline
BS−6 months vs. baseline
Veillonella
Akkermansia
Klebsiella
Actinomyces
Blautia
Ruminiclostridium_5
Spearman coefficient = 0.568
Adjusted p = 1.78e−08
RYGB_SaoPaulo−T2 vs. baseline
Assal−3 months vs. baseline
Veillonella
Streptococcus
Fusobacterium
Escherichia/Shigella
Granulicatella
Actinomyces
Klebsiella
Haemophilus
Anaerotruncus
Rothia
Gemella
Akkermansia
Fusicatenibacter
Faecalibacterium
Spearman coefficient = 0.517
Adjusted p = 4.36e−07
RYGB_SaoPaulo−T3 vs. baseline
Assal−3 months vs. baseline
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Klebsiella
Rothia
Haemophilus
Anaerotruncus
Granulicatella
Leptotrichia
Gemella
Faecalibacterium
Fusicatenibacter
Spearman coefficient = 0.509
Adjusted p = 2.44e−10
Lazaro_GB−Post surgery vs. baseline
Assal−3 months vs. baseline
Veillonella
Akkermansia
Campylobacter
Escherichia/Shigella
Anaerotruncus
Granulicatella
Klebsiella
Actinomyces
Turicibacter
Anaerostipes


Spearman coefficient = 0.478
Adjusted p = 3.74e−06
Veillonella
Streptococcus
Fusobacterium
Escherichia/Shigella
Haemophilus
Methanobrevibacter
Ruminococcaceae_UCG−003
Akkermansia
Blautia
RYGB_SaoPaulo−T2 vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.549
Adjusted p = 6.28e−08
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Haemophilus
Ruminococcaceae_UCG−003
Catenibacterium
Methanobrevibacter
RYGB_SaoPaulo−T3 vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.511
Adjusted p = 1.78e−10
Veillonella
Akkermansia
Escherichia/Shigella
Turicibacter
Anaerostipes
Blautia
Oscillibacter
Lazaro_GB−Post surgery vs. baseline
Assal−1 year vs. baseline
Spearman coefficient = 0.451
Adjusted p = 1.44e−05
Veillonella
Streptococcus
Fusobacterium
Escherichia/Shigella
Granulicatella
Haemophilus
Gemella
Shuttleworthia
Blautia
RYGB_SaoPaulo−T2 vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.565
Adjusted p = 2.12e−08
Veillonella
Streptococcus
Escherichia/Shigella
Fusobacterium
Haemophilus
Granulicatella
Lachnospiraceae_UCG−010
Leptotrichia
Gemella
RYGB_SaoPaulo−T3 vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.502
Adjusted p = 5.05e−10
Veillonella
Escherichia/Shigella
Granulicatella
Anaerostipes
Blautia
Oscillibacter
Lazaro_GB−Post surgery vs. baseline
Assal−2 years vs. baseline
Spearman coefficient = 0.384
Adjusted p = 0.00102
Veillonella
Fusobacterium
Akkermansia
UBA1819
RYGB_SaoPaulo−T2 vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.333
Adjusted p = 0.00475
Veillonella
Fusobacterium
RYGB_SaoPaulo−T3 vs. baseline
Ilhan−6 months vs. baseline
Spearman coefficient = 0.375
Adjusted p = 5.78e−05
Veillonella
Akkermansia
Lazaro_GB−Post surgery vs. baseline
Ilhan−6 months vs. baseline


Spearman coefficient = 0.261
Adjusted p = 0.0279
Veillonella
Escherichia/Shigella
Haemophilus
Blautia
RYGB_SaoPaulo−T2 vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.215
Adjusted p = 0.0718
Veillonella
Escherichia/Shigella
Haemophilus
RYGB_SaoPaulo−T3 vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.166
Adjusted p = 0.0812
Veillonella
Escherichia/Shigella
Blautia
Lazaro_GB−Post surgery vs. baseline
Ilhan−1 year vs. baseline
Spearman coefficient = 0.477
Adjusted p = 1.92e−05
Veillonella
Streptococcus
Granulicatella
Gemella
Actinomyces
Anaerotruncus
Rothia
Akkermansia
Ruminococcaceae_UCG−010
Faecalibacterium
RYGB_SaoPaulo−T2 vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = 0.455
Adjusted p = 5.02e−05
Veillonella
Streptococcus
Anaerotruncus
Rothia
Ruminococcaceae_UCG−002
Ruminococcaceae_NK4A214_group
Granulicatella
Gemella
Faecalibacterium
RYGB_SaoPaulo−T3 vs. baseline
Afshar−Post surgery vs. baseline
Spearman coefficient = 0.396
Adjusted p = 1.98e−05
Veillonella
Akkermansia
Granulicatella
Anaerotruncus
Actinomyces
Lazaro_GB−Post surgery vs. baseline
Afshar−Post surgery vs. baseline

Phase 3 overall

OG + all 4 phase−3 cohorts together

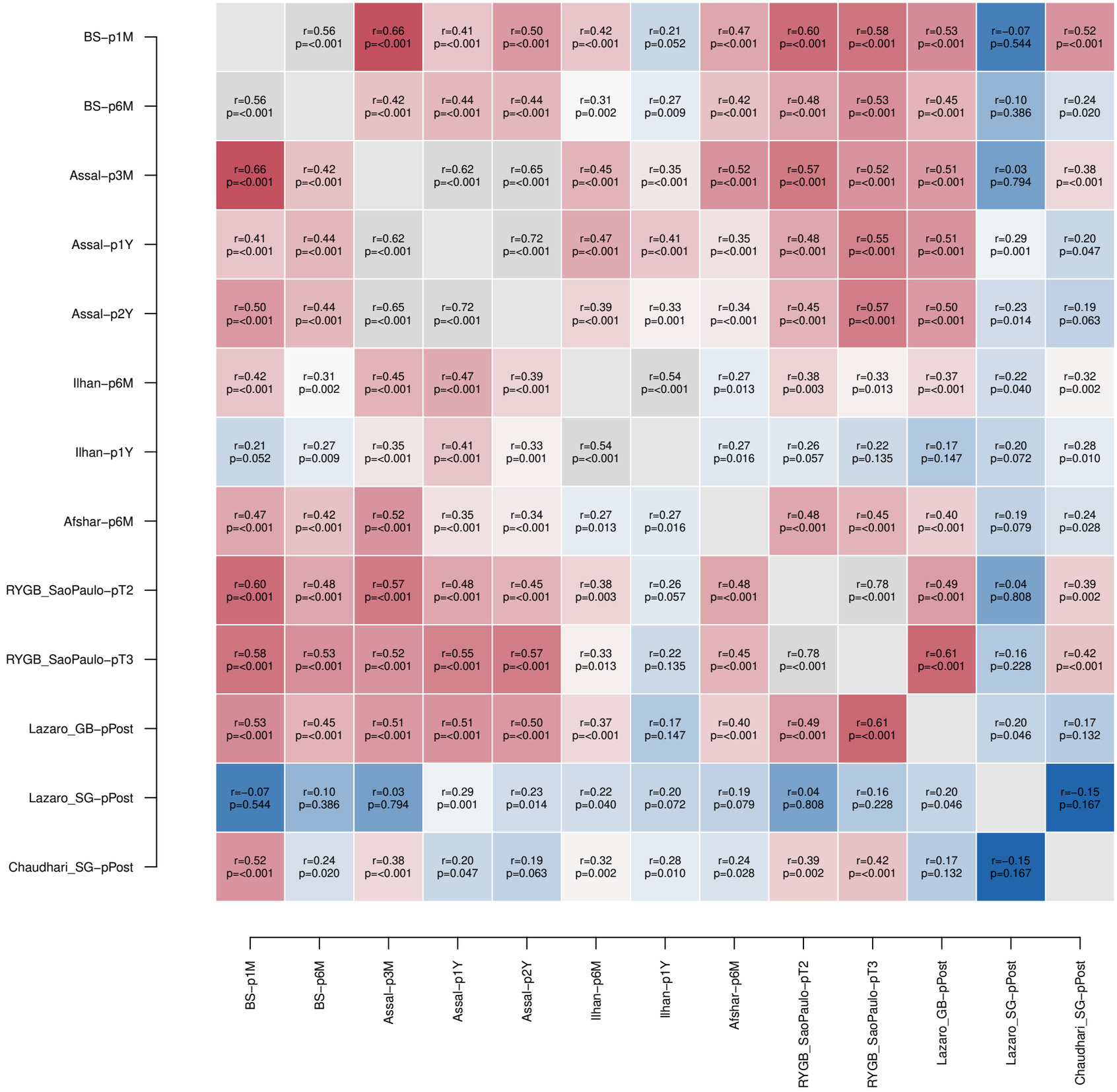
(RYGB_SaoPaulo, Lazaro_GB, Lazaro_SG, Chaudhari_SG)

Heatmap of all pairwise Spearman r, including 3a−vs−3b pairs

never computed in the separate 3a/3b runs

Pairwise Spearman r (signed −log10 p) between study−timepoints

Cell text: r and BH−adjusted p. Grey = same study (excluded from headline r). Colour: −0.15 (blue) to 0.78 (red)



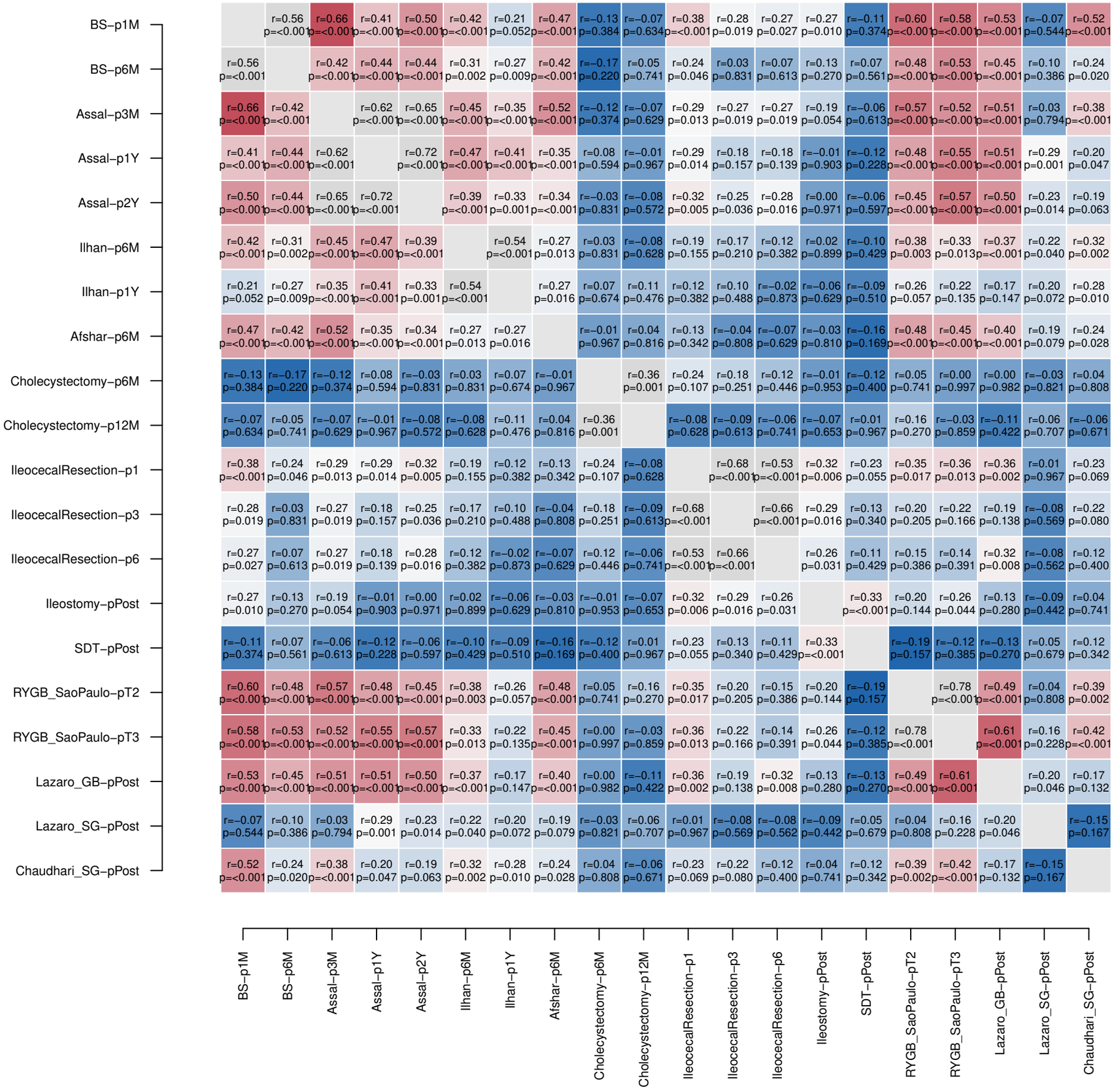
All cohorts so far

OG + Phase 2 (4) + Phase 3 (4) = 12 cohorts total

Heatmap of all pairwise Spearman r across every cohort in the project

Pairwise Spearman r (signed −log10 p) between study−timepoints

Cell text: r and BH−adjusted p. Grey = same study (excluded from headline r). Colour: −0.19 (blue) to 0.78 (red)



Top Movers and Antibiotic Cross–Reference

Which taxa increase most post–surgery across all 12 cohorts,

and whether any dataset's antibiotic use could explain it

Top Movers: Which Taxa Increase Most Post–Surgery

Largest single significant increases (BH–adjusted p < 0.05), any cohort/timepoint

Taxon	Cohort (group / timepoint)	Effect size (s)	Adj. p
Veillonella	RYGB_SaoPaulo (3b / T2)	1.97	1.3e−10
Veillonella	RYGB_SaoPaulo (3b / T3)	1.87	1.4e−10
Klebsiella	BS (OG / 1M)	1.85	3.7e−07
Fusobacterium	RYGB_SaoPaulo (3b / T2)	1.70	2.3e−05
Veillonella	Lazaro_GB (3b / Post)	1.63	4.1e−02
Klebsiella	BS (OG / 6M)	1.60	3.6e−04
Marvinbryantia	Cholecystectomy (Ph2 / 12M)	1.51	9.4e−03
Streptococcus	RYGB_SaoPaulo (3b / T3)	1.51	7.2e−07
Streptococcus	RYGB_SaoPaulo (3b / T2)	1.51	2.0e−06
Escherichia/Shigella	RYGB_SaoPaulo (3b / T3)	1.48	1.1e−04
Klebsiella	RYGB_SaoPaulo (3b / T3)	1.43	2.1e−03
Veillonella	BS (OG / 1M)	1.24	9.7e−06

s = model coefficient for that timepoint vs. baseline (positive = increased abundance post–surgery)

Most Consistently Increased Taxa Across Cohorts

Ranked by number of cohort–timepoints with a significant increase

Taxon	n sig.	Groups present in	Mean s
Veillonella	10	OG, Phase2, Phase3b	1.26
Streptococcus	6	OG, Phase3b (RYGB only)	1.11
Rothia	4	OG, Phase3b (RYGB only)	0.85
Fusobacterium	4	OG, Phase3b (RYGB only)	1.19
Granulicatella	4	OG, Phase3b (RYGB only)	0.76
Actinomyces	4	OG, Phase3b (RYGB only)	0.65
Haemophilus	4	OG, Phase3b (RYGB only)	0.89
Anaeroglobus	4	OG, Phase2	0.70
Klebsiella	3	OG, Phase3b (RYGB only)	1.63
Escherichia/Shigella	3	OG, Phase3b (RYGB only)	1.29

"(RYGB only)" = significantly increased in OG and/or Phase 3b cohorts,
never significantly increased in any Phase 2 or Phase 3a cohort.

Antibiotic Cross-Reference

Does any dataset's antibiotic use plausibly explain the Veillonella/Streptococcus rise?

Per-dataset antibiotic use (from each source paper; see project notes for full detail):
  BS (Fouladi, this study): prophylactic antibiotics routinely given before surgery; drug/dose not named in the paper.
  Assal, Afshar, Ilhan, RYGB_SaoPaulo, Chaudhari_SG: not reported or not accessible.
  Lazaro_GB and Lazaro_SG: explicitly NO antibiotic prophylaxis used.

Key finding: Veillonella rises significantly in Lazaro_GB (s=1.63, adj.p=0.041) -- a cohort with NO antibiotic exposure at all. The same taxon that dominates the 'top movers' list in antibiotic-exposed cohorts also rises in a cohort with none. This argues AGAINST antibiotics being the primary driver of the Veillonella bloom -- it looks like a genuine post-surgical effect (altered gastric acid, bile flow, and oxygen gradient), not purely an antibiotic-selection artifact.

Caveat: BS is the one cohort with CONFIRMED antibiotic use and shows the strongest Klebsiella/Veillonella/Streptococcus increases of any cohort -- antibiotic contribution there specifically cannot be ruled out, since the drug was never named. The original paper's own Discussion raises exactly this possibility ('routinely administered operative prophylactic antibiotics... might be the reason for the increase in opportunistic pathogens after surgery').

General pharmacology context (not a confirmed per-study match -- no paper names a specific drug): standard surgical prophylaxis (e.g. a first-generation cephalosporin such as cefazolin) is active against gram-positive cocci including Streptococcus, and sometimes some gram-negatives; anaerobic coverage (e.g. metronidazole) would target Fusobacterium. These are plausible mechanisms, not verified claims -- none of the 12 source papers reports the specific antibiotic used.